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A general Physics-Informed Neural Network approach for deriving fluid flow fields from temperature distribution

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ABSTRACT

One attractive application of Physics-Informed Neural Network (PINN) is deriving fluid flow information such as velocity and pressure from temperature field. However, this requires developing a suitable loss function for a given flow condition. In this paper we implemented a general loss function in PINN to directly obtain fluid velocity and pressure from temperature information. We first validated this method through cases of natural convection in a square cavity, flow around a cylinder in a square cavity, and two-dimensional channel flow. Additionally, we examined forced convection heat transfer around a cylinder in depth. Results show that different Reynolds numbers (2, 10, 40, and 140) can all be effectively resolved using temperature data. Furthermore, super-resolution predictions of temperature, velocity and pressure are achievable even far outside the training area, suggesting that the proposed PINN method could be used to measure fluid flow within confined spaces via finite visualization windows without tracers.

1. Introduction

Understanding the flow dynamics of fluid flows in chemical reaction processes requires detailed knowledge of velocity, temperature and pressure fields, and has been the focus for both experimental and theoretical fluid mechanics. Common experimental methods include intrusive measurement such as temperature or pressure probe point measurement technology (Rahman et al. 2017; van Ommen and Mudde 2008). There are also non-invasive image visualizations including electrical capacitance tomography (Wang et al. 2006a, b), ultrafast magnetic resonance (Muller et al. 2006) and particle image velocimetry (Schroder and Schanz 2023). Among them, so far the most accurate fluid velocity measurement technique is still particle image velocimetry (PIV) which relies on the tracing of small tracer particles.

The basic configuration of a PIV system includes an optically transparent test-section, an illuminating light source (laser), a recording medium (film, CCD, or holographic plate), and a computer for image processing (Prasad 2000). Many researchers use PIV to study dynamic flow behavior of bubbles in gas–solid fluidized bed (Lewandowski et al. 2019; Ziegenhein et al. 2016). But in PIV, the temporal and spatial resolutions are still constraints (Neogi et al. 2023). In addition, the tracer particles need to be carefully chosen such that the light scattering intensity should be sufficient.

In recent years machine learning algorithms have been widely used in natural language processing, speech processing, image recognition and other fields. Using the input from the sample generator and observations from the system to generate an approximation of its output, it has the characteristic of high fit and strong generalization ability, and is also making rapid progress in the field of fluid mechanics, providing a modular and agile modeling framework (Brunton et al. 2020).

Flow modeling based on machine learning contains data-driven modeling and physical mechanism modeling. Data-driven modeling includes: dimensionality reduction (Sirovich 1987; Sirovich and Kirby 1987), clustering (Kaiser et al. 2014) and classification (Colabrese et al. 2018), sparse (Manohar et al. 2018) and stochastic methods (Halko et al. 2011), and super-resolution (Dong et al. 2014; Lee et al. 2017; Xie et al. 2018). Modeling based on physical mechanism includes: linear models via nonlinear embedding (Mezic 2013; Schmid 2010), neural network modeling (Raissi and Karniadakis 2018), parsimonious nonlinear models (Schmidt and Lipson 2009), and closed models for machine learning (Ling et al. 2016a, b).

Encoding physical equations into learning algorithms, the handling of nonlinear problems introduces two important constraints (linearization and prior assumptions) (Raissi et al., 2019a). Compared to

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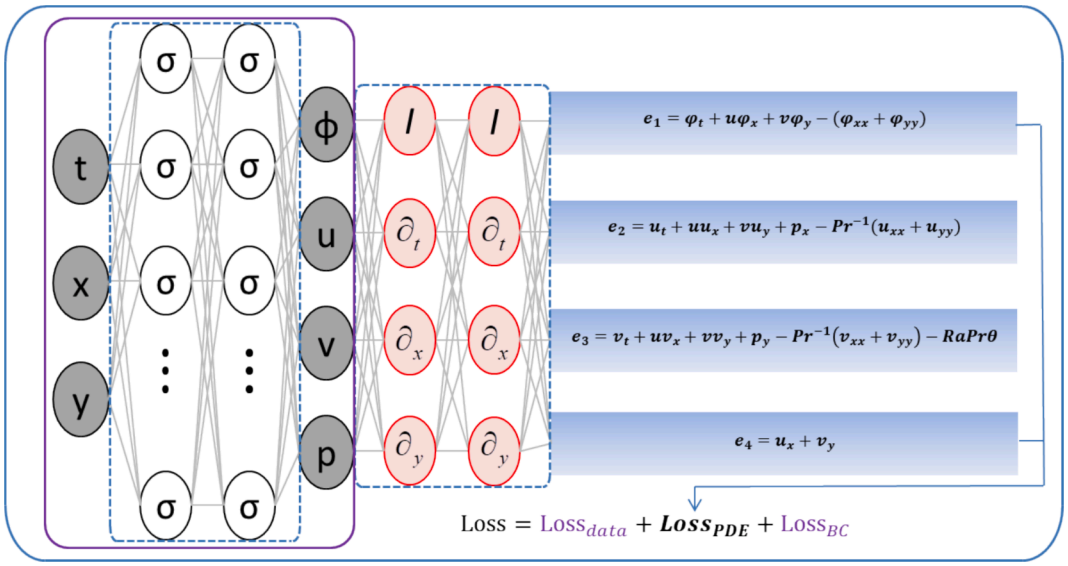

Fig. 1. A physics-uninformed neural network and physics-informed neural network for natural convection in a square cavity and flow around a cylinder in a square cavity (the purple solid line rectangle represents physics-uninformed neural network and the blue solid line rectangle represents physics-informed neural network).

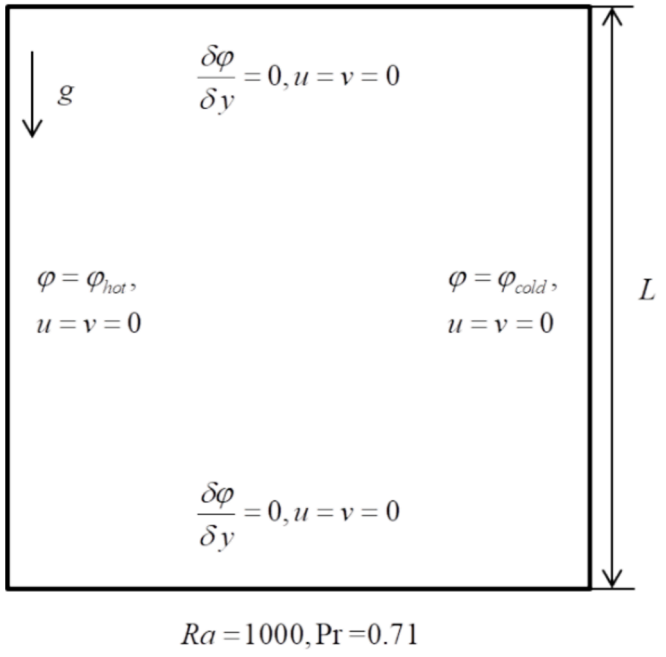

Fig. 2. Physical model of the natural convection in a square cavity.

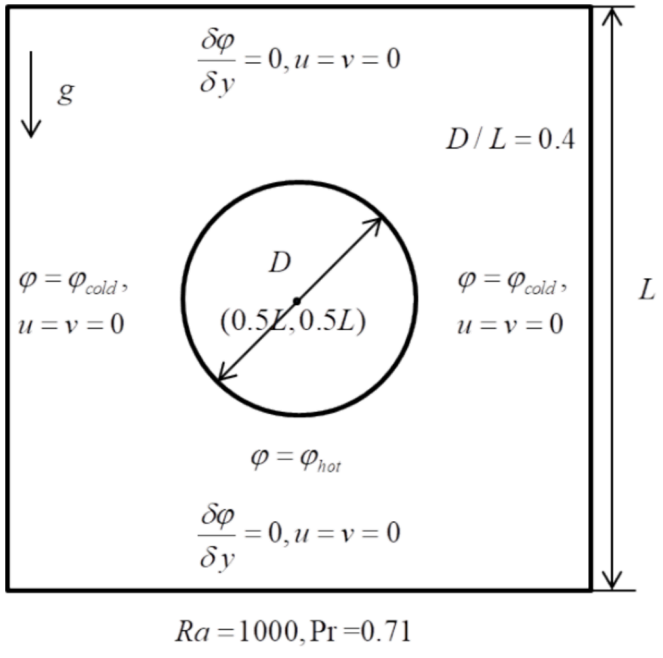

Fig. 3. Physical model of the flow around a cylinder in a square cavity.

data–driven modeling, physics-based neural networks must obey symmetry, invariance or conservation laws derived from physical laws (Karniadakis et al. 2021). Physics-informed neural network (PINN) embeds partial differential equations into the neural network loss function for the first time using auto differentiation, making the prediction results interpretable (Raissi et al., 2019a). This method can be applied to the data-driven solution of partial differential equations, enabling the extraction of fluid information that conforms to physical laws from a large amount of data. When the training loss for PINN includes terms for the PDEs, boundary conditions, and initial conditions, it can be regarded as solving forward problem. IB-PINN is a novel PINN that incorporates the direct-forcing immersed boundary method. Its loss function consists of partial differential equations, boundary conditions, immersion boundaries and force error. This unique approach can directly calculate flow field velocity, pressure and cylindrical force of

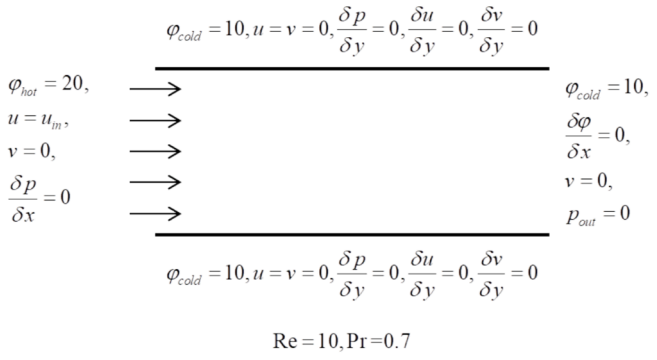

Fig. 4. Physical model of the two dimensional channel flow.

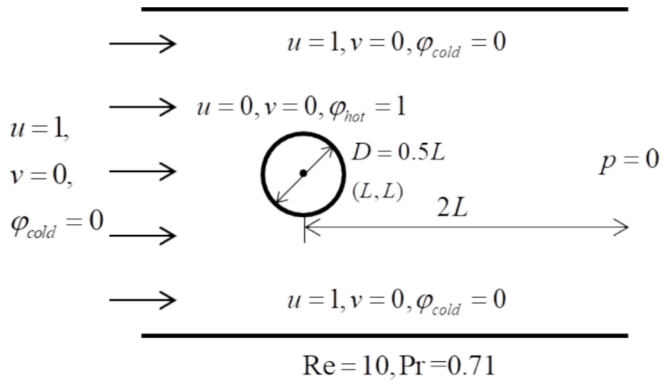

Fig. 5. Physical model of forced convection heat transfer around a cylinder.

Table 1
Spatial resolution, time step interval, training snapshots, and final loss for the trained data at all cases.

Model	Spatial resolution	Time step interval	Training snapshots	Final loss
Natural convection in a square cavity	800 × 800	1000	201	2.4×10^{-2}
Flow around a cylinder in a square cavity	800 × 800	1000	101	8.6×10^{-2}
Two dimensional channel flow	1600 × 1600	1000	201	4.2×10^{-4}
Forced convection heat transfer around a cylinder (Re = 2)	300 × 200	1	28	1.2×10^{-4}
Forced convection heat transfer around a cylinder (Re = 10)	300 × 200	1	28	3.2×10^{-4}
Forced convection heat transfer around a cylinder (Re = 40)	300 × 200	1	28	1.7×10^{-3}
Forced convection heat transfer around a cylinder (Re = 140)	600 × 400	1	28	1.6×10^{-3}

laminar flow past a circular cylinder that is placed in a channel (Huang et al. 2022). Extended PINN (XPINN) allows the solution of differential equations through space–time decomposition, achieved by incorporating PINN across multiple regions. The loss function of XPINN comprises both the initial conditions and partial differential equations of each region and junction. The entire field velocity can be predicted accurately by the boundary velocity and local velocity of the flow filed

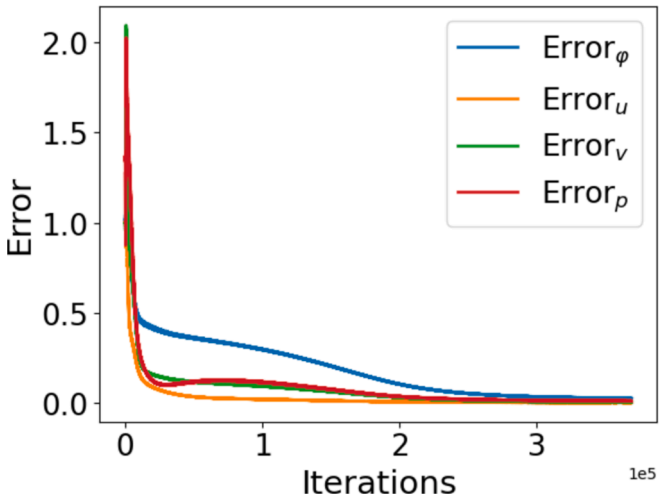

Fig. 7. Relative error estimated for φ, p, u, v of natural convection in a square cavity.

(Shukla et al. 2021). But the above researches only focus on Navier-Stokes (NS) equation not including energy equation. When observed data loss is added to the loss function, the PINN can be used to solve inverse problems. Recently hidden fluid mechanics (HFM) applies automatic differentiation on the output variables, encodes the transport and NS equations in the physics informed neural network, and finally learns the shared parameters of the physics uninformed neural network and physics informed neural network by minimizing the loss function. It can obtain the velocity and pressure of the flow field through the concentration of the flow field (Raissi et al. 2020). Inspired by this, PINN may be possible to be used to derive fluid flow velocity from temperature distribution.

At present, many researchers use PINN to solve heat transfer problems (Bararnia and Esmaeilpour 2022; Cai et al. 2021a, b, c; Laubscher 2021; Laubscher and Rousseau 2022; Lu et al. 2021; Manavi et al. 2023). Manazi and Lu imposed the boundary conditions of PINN with hard constraints composed by boundary function and distance function and applied the enhanced surrogate model to solve heat conduction equations, but which don't include momentum equation (Lu et al. 2021; Manavi et al. 2023). Laubscher adopted PINN to predict momentum, species and temperature distributions of a dry air humidification problem in a simple two-dimensional rectangular domain (Laubscher 2021) and incompressible laminar flow with heat transfer through a 2D internal domain with obstructions (Laubscher and Rousseau 2022). Barnia used PINN to solve boundary layer thermal fluid problems

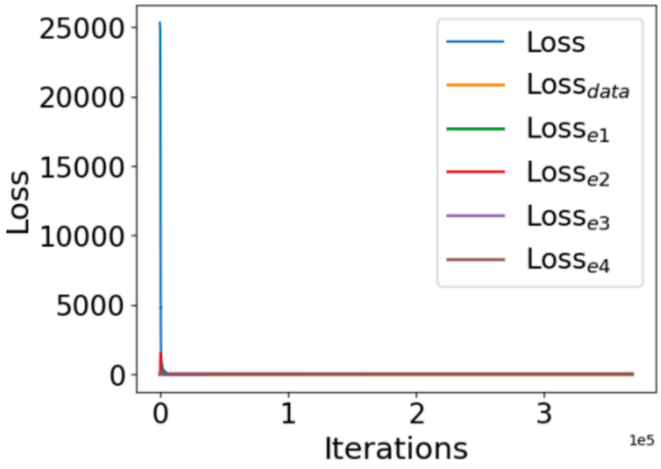

Fig. 6. Convergence of loss function of natural convection in a square cavity.

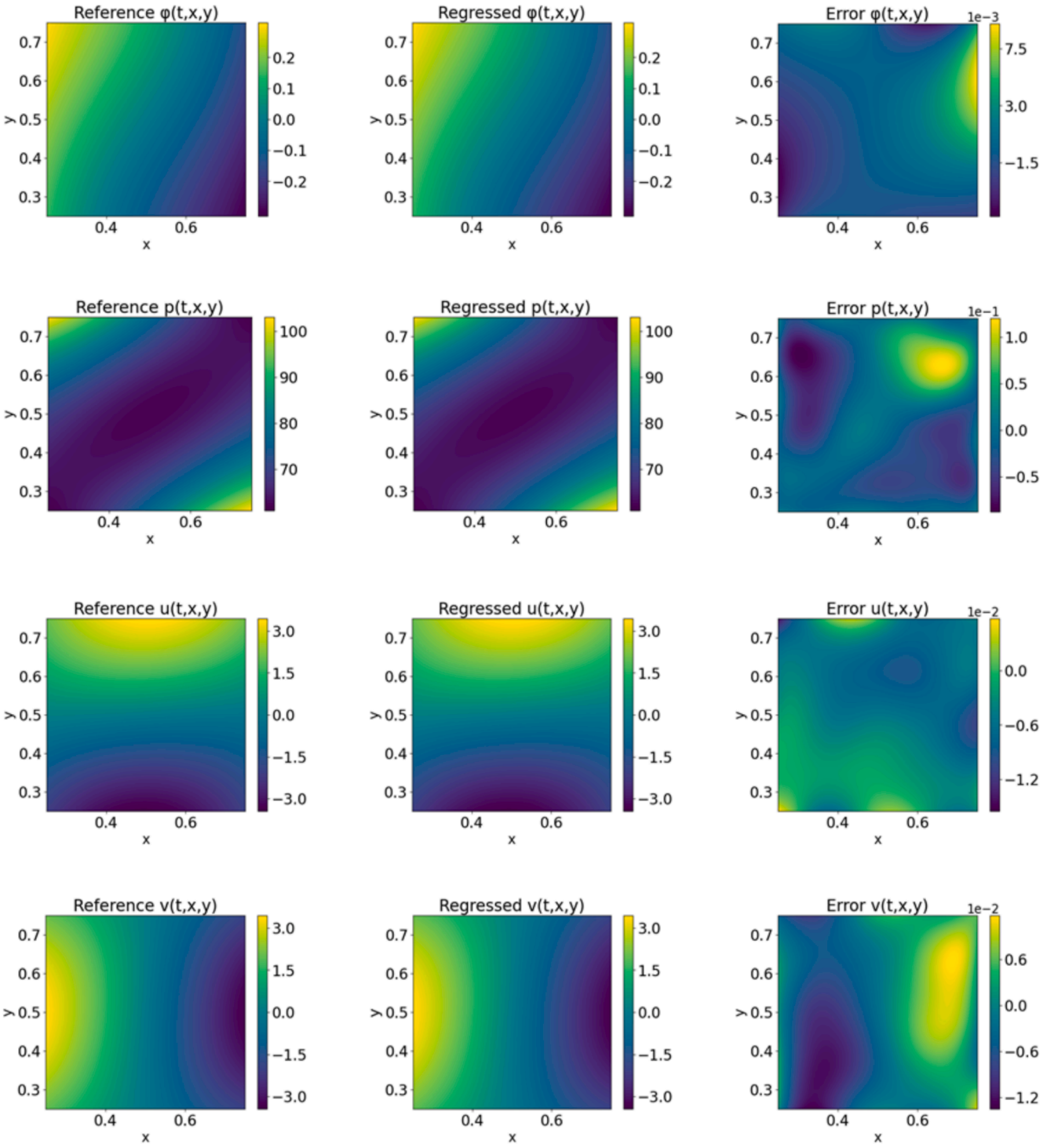

Fig. 8. Referenced and regressed φ, p, u, v, along with the corresponding absolute error for a specific selected region of natural convection in a square cavity.

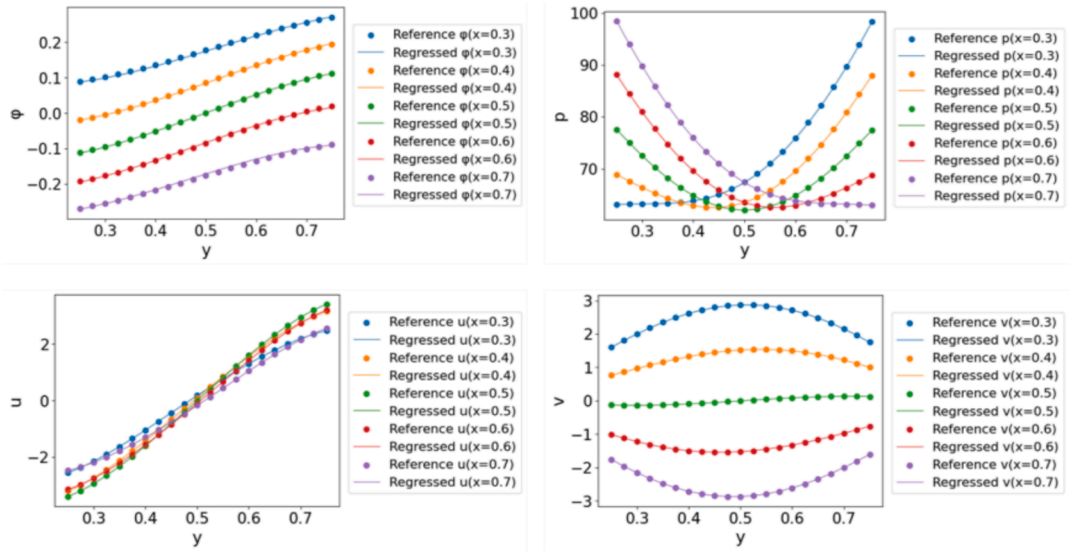

Fig. 9. Referenced and regressed φ, p, u, v for x = 0.3, 0.4, 0.5, 0.6 and 0.7 of natural convection in a square cavity.

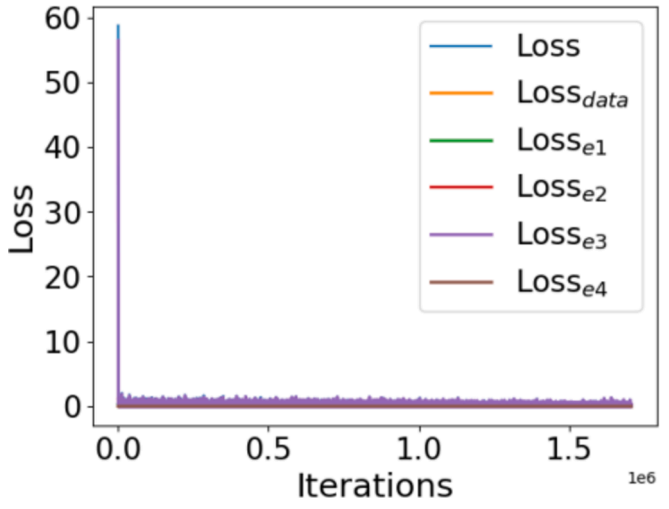

Fig. 10. Convergence of loss function of flow around a cylinder in a square cavity.

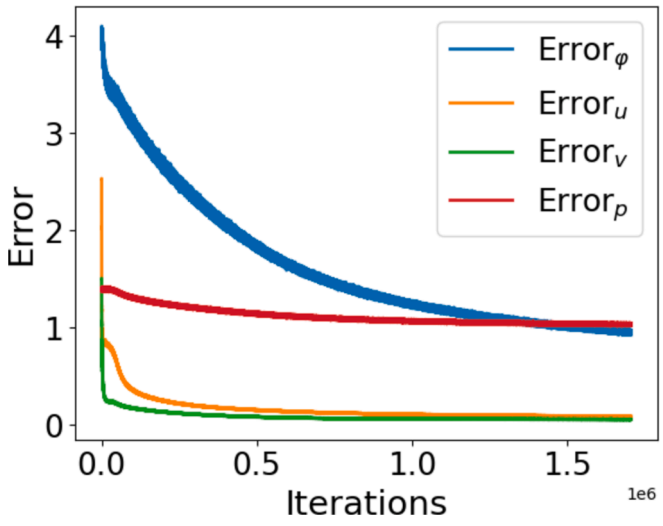

Fig. 11. Relative error estimated for φ, p, u, v of flow around a cylinder in a square cavity.

including Blasius-Pohlhausen, Falkner-Skan, and Natural convection (Bararnia and Esmaeilpour 2022). These researches encompassed not only momentum equations but also energy equations. However, higher order terms of the NS and energy equation were either reduced or even simplified into a set of coupled ordinary differential equations.

The aim of this paper is also to apply PINN to derive velocity and pressure in heat transfer processes because of the difficulty of obtaining velocity. But we solve NS equation and energy equation directly without order reduction and simplification. The composition of the loss function varies depending on the problem. Although Cai proposed a new PINN method similar to HFM to infer the full continuous three-dimensional (3-D) velocity and pressure fields from snapshots of 3-D temperature fields obtained by Tomo-BOS imaging (Cai et al., 2021b) without velocity boundary conditions, the boundary of our considered cases are more complex than the open boundary verified by Cai's experiment. Moreover, unlike the loss function proposed by Cai in the numerical simulation (Cai et al., 2021c), where for the steady forced convection problem, the PINN loss function includes PDEs, the boundary conditions for the velocity and temperature fields, and temperature-observed data, for the mixed convection problem, in addition to the above four terms, a system error ensuring the symmetric nature of the temperature profile needs to be added to the loss function. In contrast, our proposed loss function consists only of only the data loss (Loss_{data}), the physics equation loss (Loss_{PDE}) and the boundary condition loss (Loss_{BC}) for the velocity and pressure fields. Our method is general and simple, with no need to change the loss function under different conditions. Furthermore, the research area can be chosen arbitrarily. We first demonstrate the method for natural convection in a square cavity where velocity and temperature interact with each other. Next, we consider the flow around a cylinder in a square cavity which poses a greater challenge than the first natural convection due to the presence of cylinder. Additionally, we study two-dimensional channel flow where velocity affects temperature but temperature does not affect velocity. Finally for forced convection heat transfer around a cylinder example, we conduct an in-depth research on different conditions of Reynolds number of 2, 10, 40 and 140 and perform super-resolution predictions for temperature, velocity and pressure within and outside the training area.

The organization of this article is as follows. First the physics-informed neural network approach for deriving fluid flow fields from

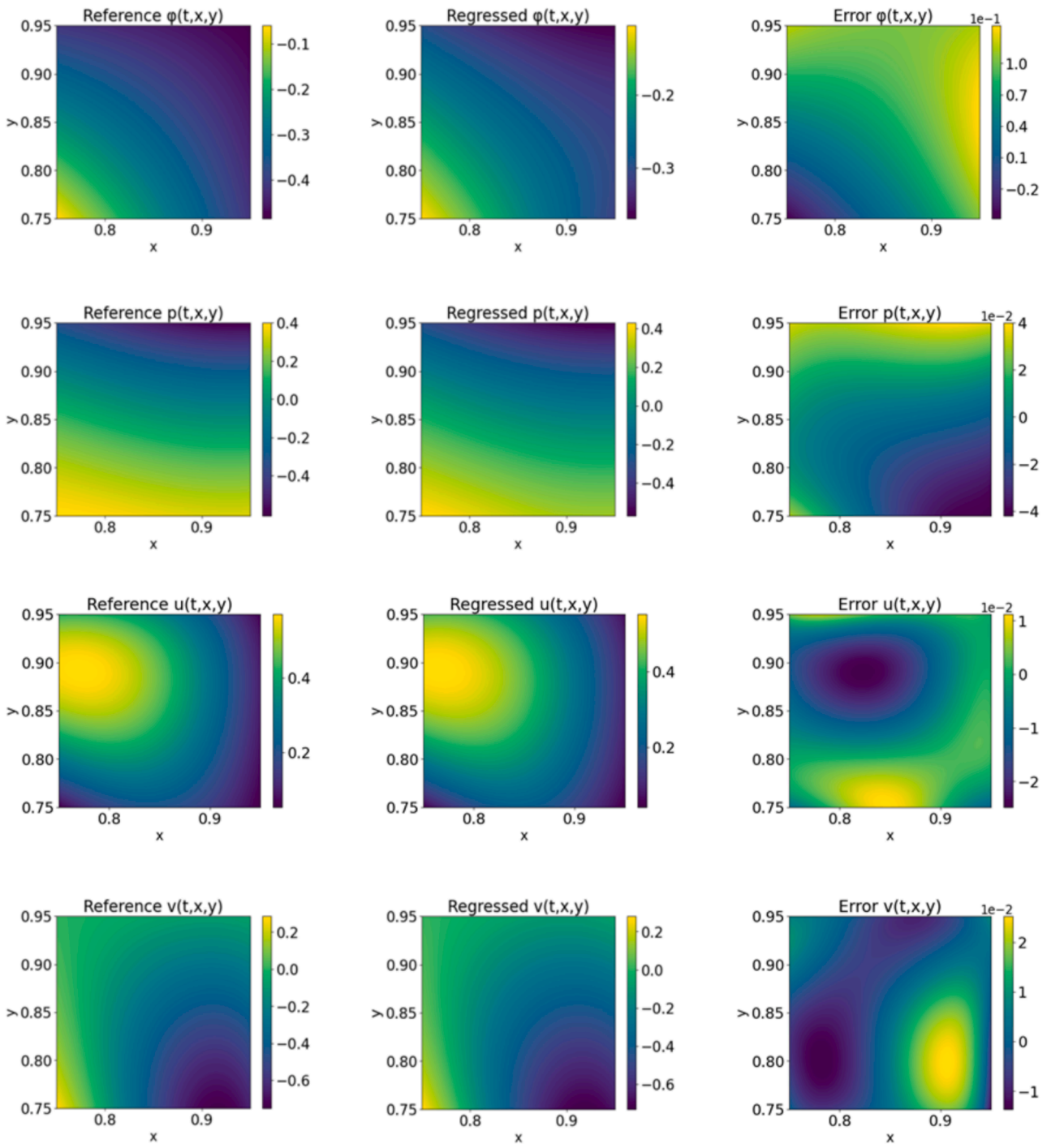

Fig. 12. Referenced and regressed φ, p, u, v, along with the corresponding absolute error for a specific selected region of flow around a cylinder in a square cavity.

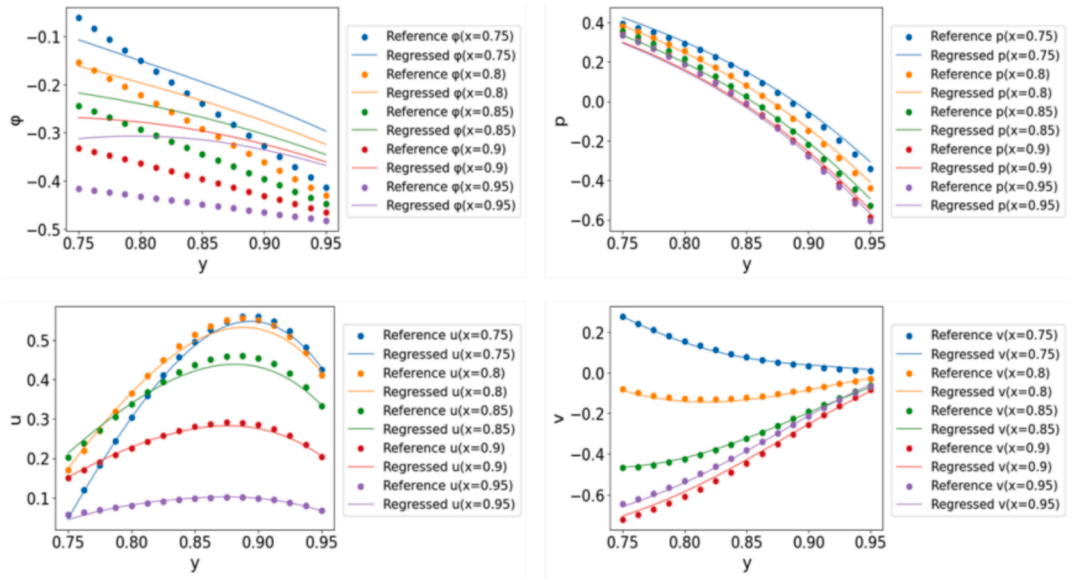

Fig. 13. Reference and Regressed φ, p, u, v for x = 0.75, 0.8, 0.85, 0.9 and 0.95 of flow around a cylinder in a square cavity.

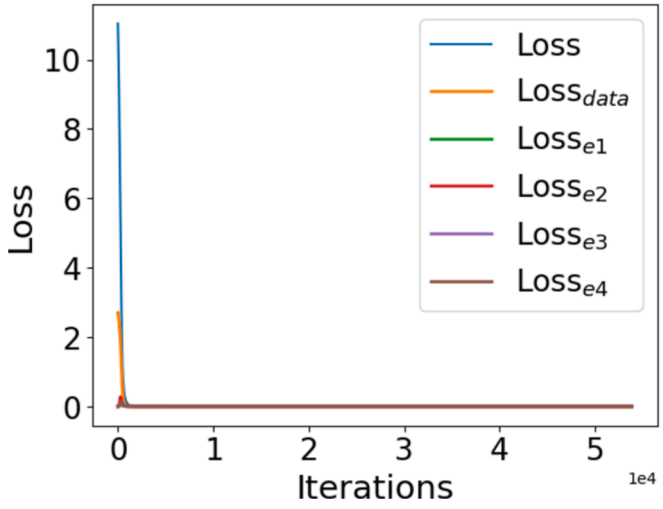

Fig. 14. Convergence of loss function of two dimensional channel flow.

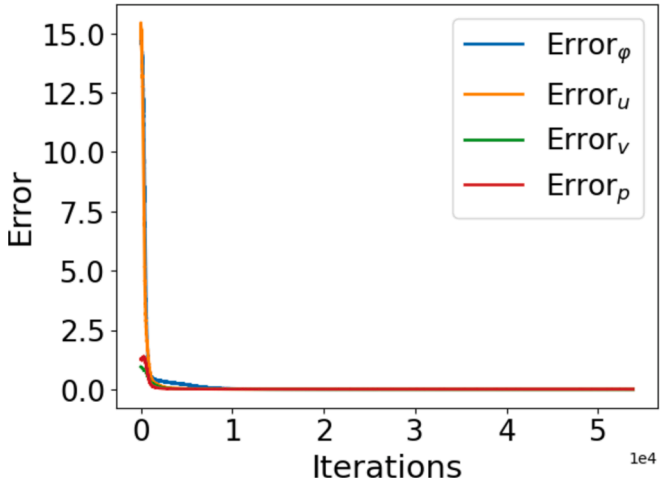

Fig. 15. Relative L_2 errors estimated for φ, p, u, v of two dimensional channel flow.

temperature distribution and four basic fluid mechanics examples are described in detail. Then, the respective inference results are presented and discussed. Finally the paper ends with some conclusions.

2. Physics-Informed neural network (PINN)

The following is the form of the partial differential equations commonly used to describe physical fields (Cai et al., 2021a):

$$f(x, t, \widehat{u}, \partial_x \widehat{u}, \partial_t \widehat{u}, \cdots, \lambda) = 0, x \in \Omega, t \in [0, T] \tag{1}$$

$$\widehat{u}(x, t_0) = m_0(x), x \in \Omega \tag{2}$$

$$\widehat{u}(x, t) = m_\Gamma(t), x \in \partial\Omega, t \in [0, T] \tag{3}$$

Where x is spatial coordinate system x = [X, Y] and t is time, f in Eq. (1) is the residual of the partial differential equation composed of the partial differential operator $[\partial_x \widehat{u}, \partial_t \widehat{u}]$ and the parameter $[\lambda_1, \lambda_2]$. $\widehat{u} = [\varphi, u, v, p]$ represents the solution of partial differential equation under the initial condition $m_0(x)$ and the boundary condition $m_\Gamma(t)$. Ω in Eq. (2) and $\partial\Omega$ in Eq. (3) stand for the space domain and the boundary respectively.

$$a^0 = (\mathrm{x}, \mathrm{t}), \tag{4}$$

$$a^k = \sigma\left(w^k a^{k-1} + b^k\right), 1 \le k \ll L - 1 \tag{5}$$

$$a^k = w^k a^{k-1} + b^k, k = L \tag{6}$$

Described by Eqs. (4) to (6), PINN network consists of a fully connected feed-forward neural network with multiple hidden layers used to approximate the output solution $\widehat{u} = [\varphi, u, v, p]$ of the input [x, t], where a^0 is the input of the network composed of spatial and time coordinate and a^k represents the output of the k layer. w^k and b^k are the weight and bias parameters of the k layer of the network, respectively. σ is the nonlinear activation function.

$$\mathrm{Loss} = \omega_1 \mathrm{Loss}_{PDE} + \omega_2 \mathrm{Loss}_{data} + \omega_3 \mathrm{Loss}_{IC} + \omega_4 \mathrm{Loss}_{BC} \tag{7}$$

The loss function of the PINN network in Eq. (7) comprises the residual error of the partial differential equation (Loss_{PDE}), the data error between the observed value and the network output (Loss_{data}), the initial condition error (Loss_{IC}) and boundary error (Loss_{BC}), while the loss

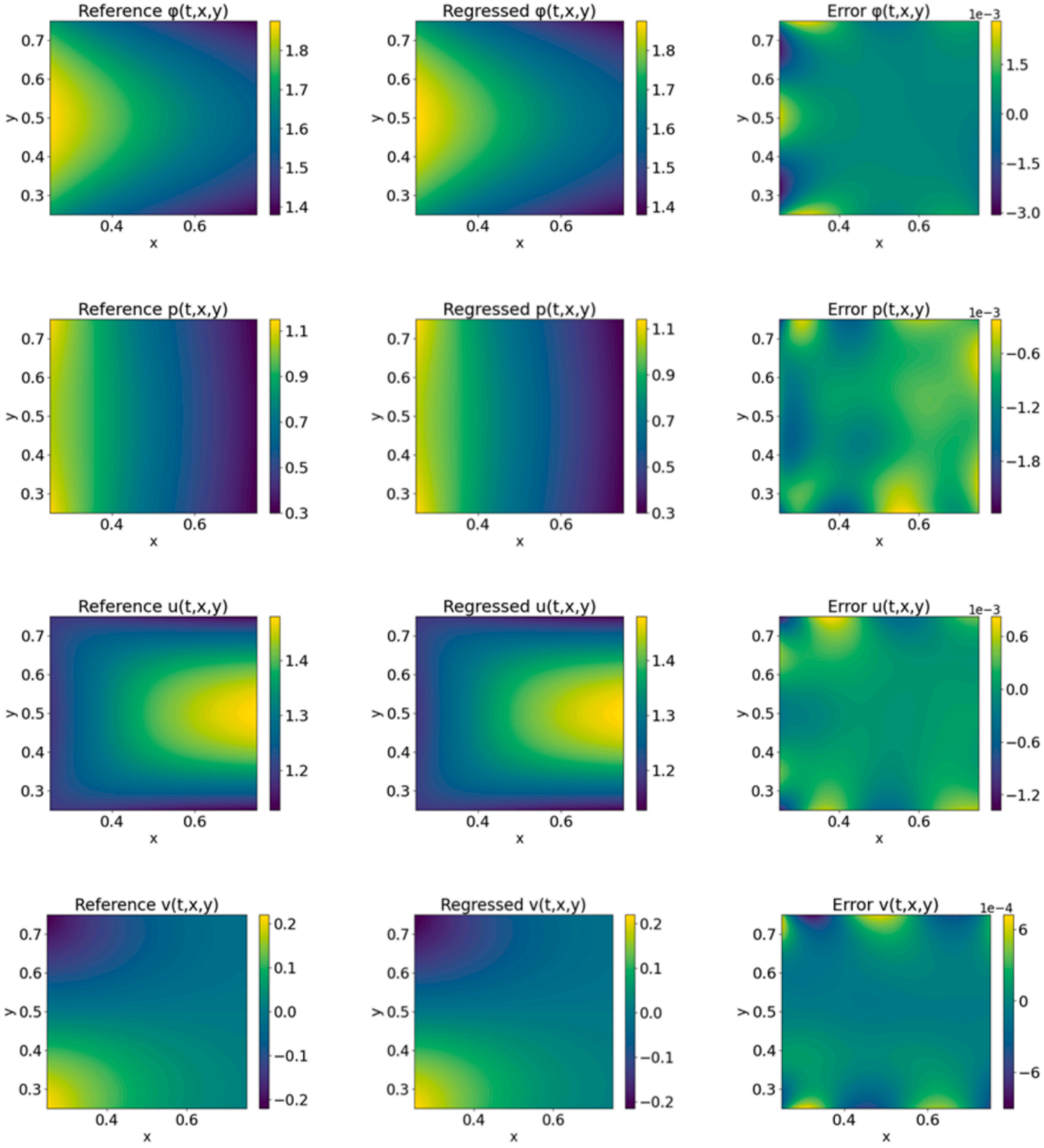

Fig. 16. Referenced and regressed φ, p, u, v, along with the corresponding absolute error for a specific selected region of two dimensional channel flow.

function of the traditional machine learning algorithm primarily focuses on the data error between the network output and the observed value. The weight $[\omega_1, \omega_2, \omega_3, \omega_4]$ is settled to 1 in this paper. Forward problems with known initial conditions, boundary conditions, and equation parameters can be solved without observations. Observations are required for inverse problems where the boundary conditions or equation parameters are unknown. The algorithm can also solve over determined system given initial boundary and observed values. In this study, we do not solve the forward problem in an unsupervised manner using PINN; instead, we solve the inverse problem by incorporating some temperature data as observed values to train the network to infer fluid velocity and pressure fields.

We set up the problems as follows: within the area of interest in the flow field, we are provided with some space–time coordinates along with corresponding temperature values from the Lattice Boltzmann direct numerical simulation method (LBM). Our objective is to compute the velocity and pressure at each given space–time coordinate. Through multiple tests, we have found that three loss terms in Eq. (8), including the data loss (Loss_{data}), the physics equation loss (Loss_{PDE}) and the boundary condition loss (Loss_{BC}) can be used to ensure the network convergence and accurate output. The loss term for the initial condition is not included since the network can correctly output temperature, velocity and pressure fields for the specified area as long as three loss terms (i.e., data loss, equation loss and boundary condition loss) are

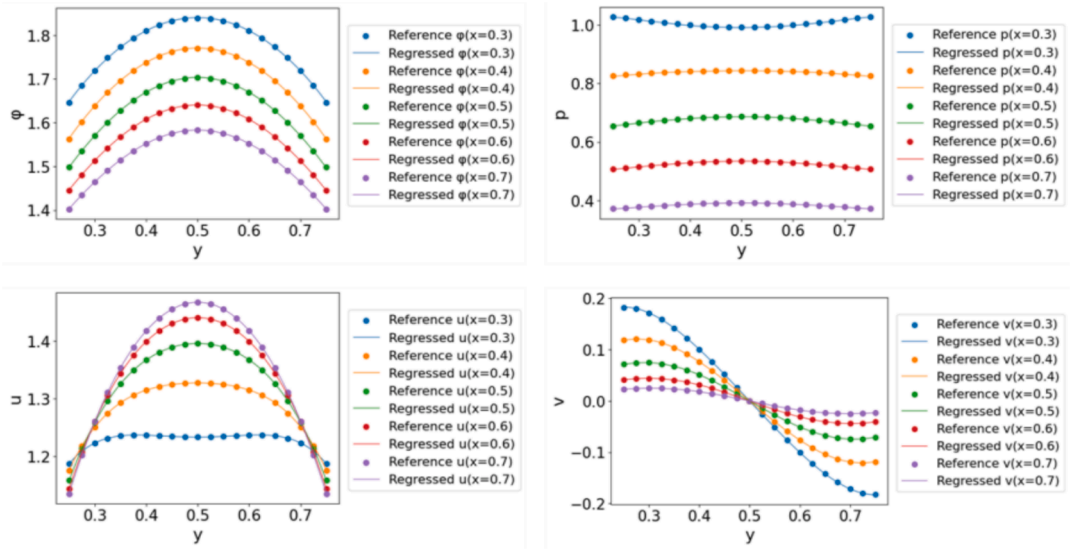

Fig. 17. Reference and Regressed φ, p, u, v for x = 0.3, 0.4, 0.5, 0.6 and 0.7 of two dimensional channel flow.

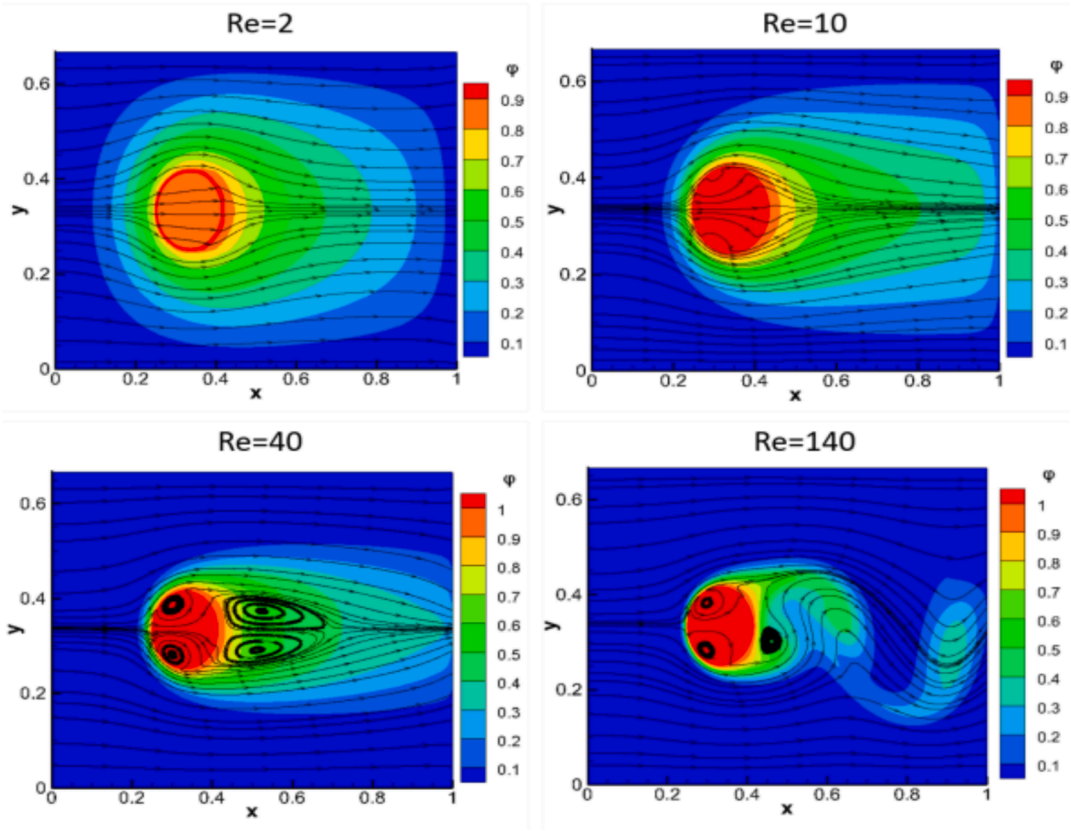

Fig. 18. Flow and temperature distribution at different Reynolds numbers of forced convection heat transfer around a cylinder (Re = 2, 10, 40 and 140).

given.

Data loss Loss_{data} in Eq. (9) aims to directly match model output temperature to the observed data, which is computed using mean squared error (MSE) between the training data and the network outputs. The physics loss Loss_{PDE} in Eq. (10) ensures that the network predictions obey known physical laws such as conservation of mass, momentum and energy. The computation process of Loss_{PDE} is as follows: first automatic differential is used to solve the partial differential operator $[\partial_x \widehat{u}, \partial_t \widehat{u}]$ based on the output of the network $\widehat{u}$. Next, the partial differential operator is applied to construct the residual of the partial differential equations. Finally Loss_{PDE} is calculated as the mean squared error (MSE) between the residual and 0.

Boundary condition loss Loss_{BC} in Eq. (11) allows the PINN to learn some information about velocity and pressure based on the boundary condition. Loss_{BC} is defined as the MSE between PINN predictions and the values at boundary condition fed into the network. HFM uses the

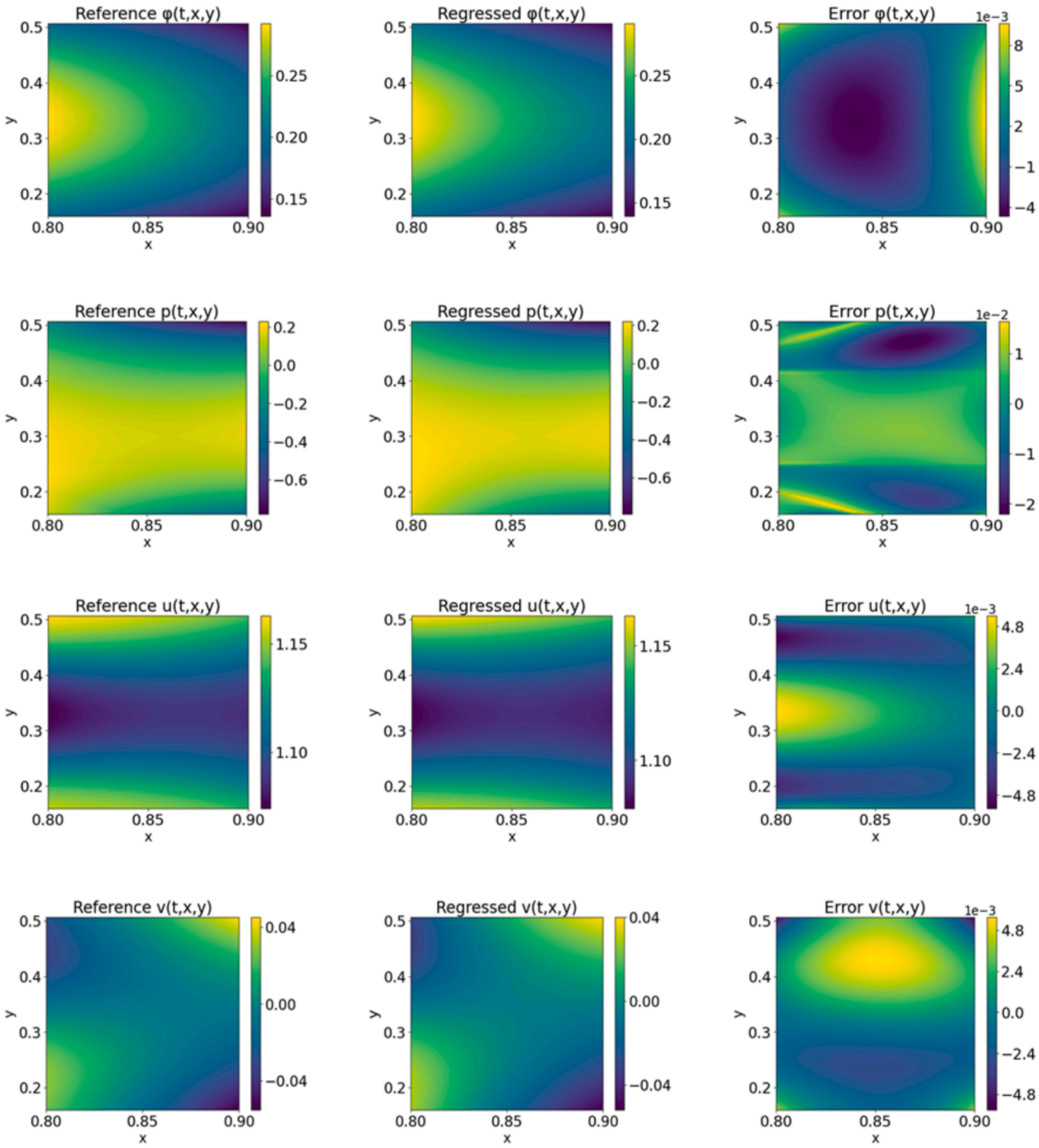

Fig. 19. Referenced and regressed φ, p, u, v, along with the corresponding absolute error for a specific selected region of forced convection heat transfer around a cylinder (Re = 2).

concentration or temperature data of the flow field to predict the velocity and pressure. In some special cases, velocity and pressure boundaries are not needed because the concentration or temperature contains sufficient boundary condition information (Cai et al., 2021b; Raissi et al., 2020). The purpose of our work is to predict the velocity and pressure of the flow field using temperature field data. However, in our validation examples, deriving velocity and pressure directly from temperature data without any information on velocity and pressure cannot be achieved. The possible reason may be that relationship between temperature and velocity (pressure) is not one-to-one but one-to-many due to the complexity of the flow. Moreover, the complexity and diversity of the loss function used by Cai in different cases verify that inference (Cai et al., 2021c). So there is need to add boundary condition loss to the loss function and our boundary condition loss Loss_{BC} consists of velocity and pressure boundary error.

Therefore the loss function Loss includes Loss_{data}, Loss_{PDE} and Loss_{BC} of u, v, p in this work is shown in Eq. (12).

$$\text{Loss} = \text{Loss}_{data} + \text{Loss}_{PDE} + \text{Loss}_{BC} \tag{8}$$

$$\text{Loss}_{data} = \frac{1}{N}\sum_{n=1}^{N} |\varphi(t^n, x^n, y^n) - \varphi^n|^2 \tag{9}$$

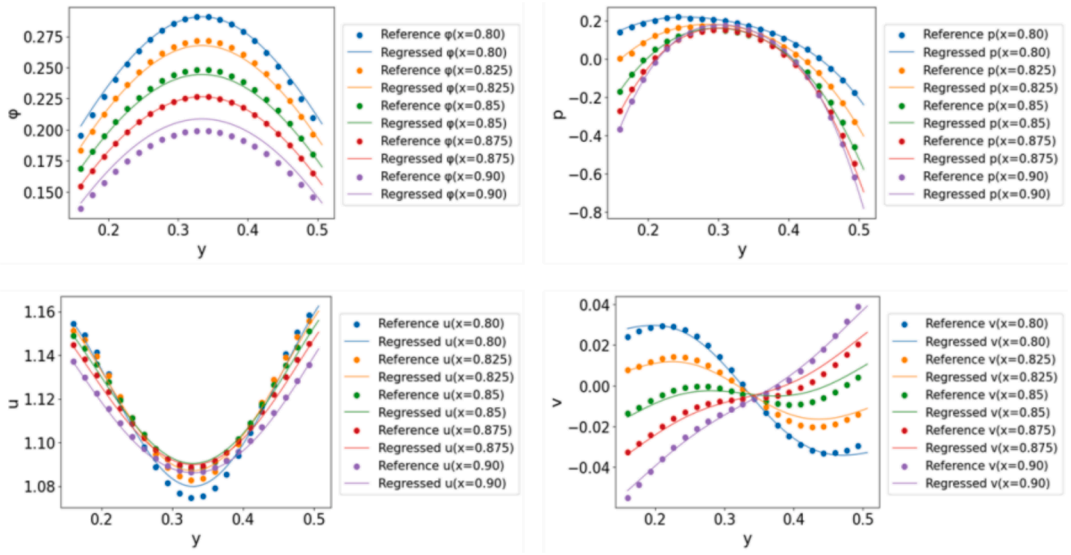

Fig. 20. Referenced and regressed φ, p, u, v for x = 0.80, 0.825, 0.85, 0.875 and 0.90 of forced convection heat transfer around a cylinder (Re = 2).

$$\text{Loss}_{PDE} = \sum_{i=1}^{4} \frac{1}{M} \sum_{m=1}^{M} |e_i(t^m, x^m, y^m)|^2 \tag{10}$$

$$\text{Loss}_{BC} = \frac{1}{K} \sum_{k=1}^{K} |\delta(t^k, x^k, y^k) - \delta^k|^2 (\delta = [u, v, p]) \tag{11}$$

$$\begin{aligned} \text{Loss} = &\frac{1}{N} \sum_{n=1}^{N} |\varphi(t^n, x^n, y^n) - \varphi^n|^2 + \sum_{i=1}^{4} \frac{1}{M} \sum_{m=1}^{M} |e_i(t^m, x^m, y^m)|^2 \\ &+ \frac{1}{K} \sum_{k=1}^{K} |\delta(t^k, x^k, y^k) - \delta^k|^2 \end{aligned} \tag{12}$$

Fig. 1 illustrates a physics-uninformed neural network and physics informed neural network for natural convection in a square cavity and flow around a cylinder in a square cavity, where fluid temperature and velocity interact with each other. A "Physics-uninformed neural network" refers to a traditional, pure data-driven network that does not embed physical equations (Raissi et al. 2020). The input to the physics-uninformed neural network consists of time and space coordinates, while the output includes temperature, velocity, and pressure. By applying automatic differentiation to the output of physics-uninformed neural network to encode physical governing equations, a physics-informed neural network is obtained. It should be noted that physics-uninformed neural network is used to construct observed data error Loss_{data} and boundary condition error Loss_{BC}. The transport and NS equations were encoded in the physics-informed neural network to get Loss_{PDE} achieved by applying automatic differentiation to the output of shared uninformed neural network. The governing equations of two-dimensional channel flow, and forced convection heat transfer around a cylinder are different from natural convection in a square cavity, and flow around a cylinder in a square cavity. Thus, only governing equations need to be changed to obtain Loss_{PDE} while the constructions of Loss_{data} and Loss_{BC} remains the same.

We utilize the TensorFlow framework to train the network. The Adam optimizer, with a learning rate of 10^{-3}, is used to optimize the network parameters $[w^k, b^k]$ so that the loss function Loss is continuously reduced and the output $\hat{u}$ closely approximates the solution of the partial differential equation.

3. Methods

3.1. Reference data generation

This work considers the heat transfer coupled fluid flows. The governing equations including continuity equation, momentum equation and energy equation are presented as Eqs. (13)–(16):

$$\frac{\delta u}{\delta x} + \frac{\delta v}{\delta y} = 0 \tag{13}$$

$$\frac{\delta u}{\delta t} + \mathrm{u}\frac{\delta u}{\delta x} + v\frac{\delta u}{\delta y} = -\frac{1}{\rho}\frac{\delta p}{\delta x} + \nu_f\left(\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2}\right) + Fx \tag{14}$$

$$\frac{\delta v}{\delta t} + \mathrm{u}\frac{\delta v}{\delta x} + v\frac{\delta v}{\delta y} = -\frac{1}{\rho}\frac{\delta p}{\delta y} + \nu_f\left(\frac{\delta^2 v}{\delta x^2} + \frac{\delta^2 v}{\delta y^2}\right) + Fy \tag{15}$$

$$\frac{\delta \varphi}{\delta t} + \mathrm{u}\frac{\delta \varphi}{\delta x} + v\frac{\delta \varphi}{\delta y} = \nu_t\left(\frac{\delta^2 \varphi}{\delta x^2} + \frac{\delta^2 \varphi}{\delta y^2}\right) \tag{16}$$

where φ, u, v, p, ρ represent temperature, the x-direction component of velocity, the y-direction component of velocity, pressure and density. Fx and Fy are body force term of x-direction and y-direction.ν_f and ν_t are kinematic viscosity and thermal conductivity respectively.

To generate reference data, the Lattice Boltzmann direct numerical simulation method is applied, where the distribution functions g in Eq. (17) and h in Eq. (18) are utilized to calculate the velocity and the temperature field respectively as:

$$\mathrm{g}_a(\boldsymbol{x} + \boldsymbol{e}_a\delta t, t + \delta t) = g_a(\boldsymbol{x}, t) - \Omega_g(g - g^{eq}) + w_a F\delta t \tag{17}$$

$$\mathrm{h}_a(\boldsymbol{x} + \boldsymbol{e}_a\delta t, t + \delta t) = h_a(\boldsymbol{x}, t) - \Omega_h(h - h^{eq}) + w_a Q\delta t \tag{18}$$

In the above Equations $\mathbf{x}$=[x, y], Ω_g and Ω_h are the collision factors related to ν_f and ν_t respectively. a represents the a^{th} direction for the lattice Boltzmann distribution functions, with the distribution parameters defined as $\mathrm{e} = [(0,0), (1,0), (0,1), (-1,0), (0,-1), (1,1), (-1,1), (-1,-1), (1,-1)]$ and $\mathrm{w} = \left[\frac{4}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{36}, \frac{1}{36}, \frac{1}{36}, \frac{1}{36}\right]$.

The superscript equations represent the equivalent station calculated by Eq. (19) and Eq. (20):

$$\mathrm{g}^{eq}(\boldsymbol{x}, t) = w_a\rho\left(1 + \frac{\boldsymbol{u}\cdot\boldsymbol{e}_a}{c_s^2} + \frac{(\boldsymbol{u}\cdot\boldsymbol{e}_a)^2}{2c_s^4} - \frac{\boldsymbol{u}\cdot\boldsymbol{u}}{2c_s^2}\right) \tag{19}$$

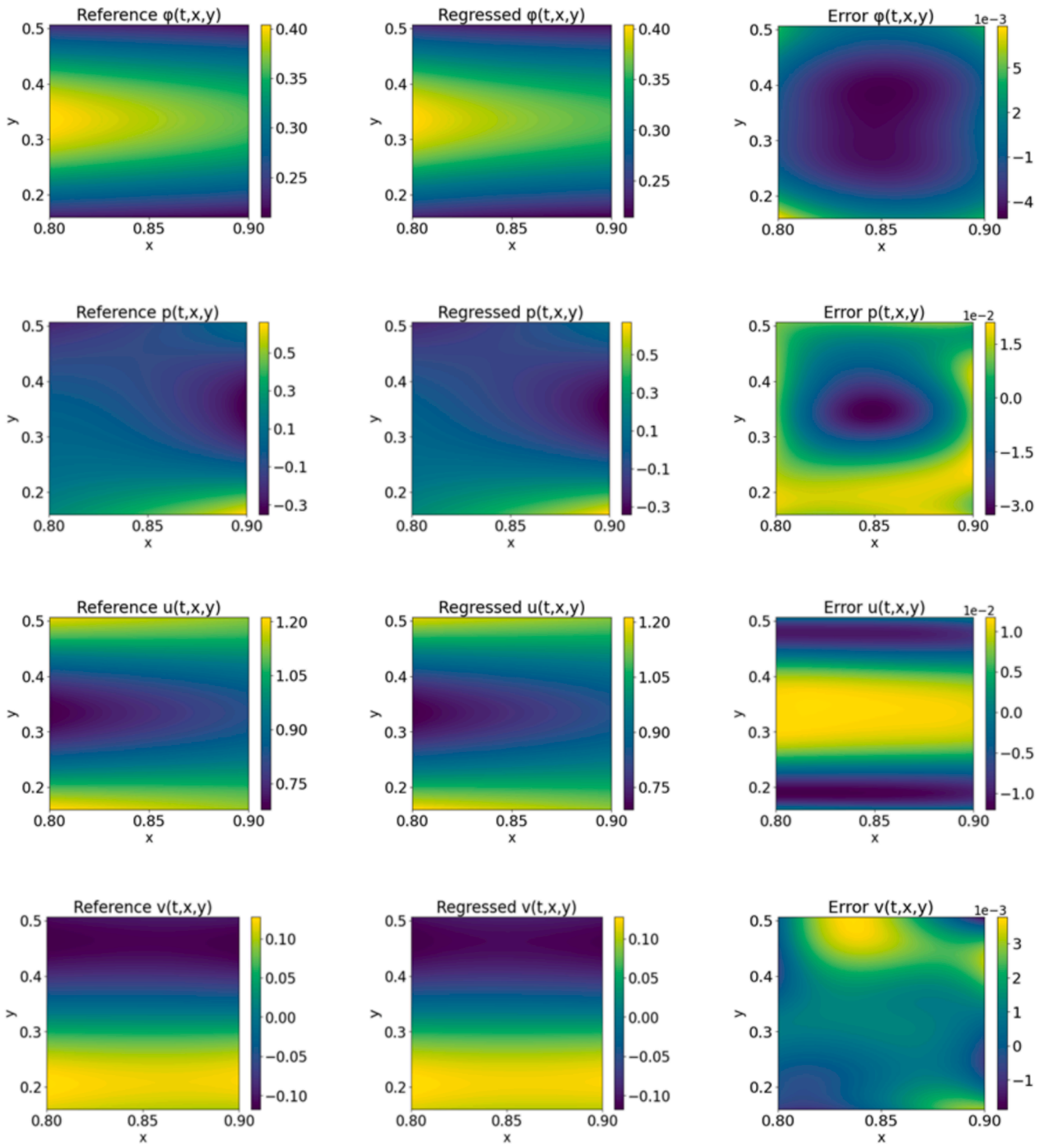

Fig. 21. Referenced and regressed φ, p, u, v, along with the corresponding absolute error for a specific selected region of forced convection heat transfer around a cylinder (Re = 10).

$$\mathrm{h}^{eq}(\boldsymbol{x},t) = w_a T\left(1+\frac{\boldsymbol{u}\cdot\boldsymbol{e}_a}{c_s^2}+\frac{(\boldsymbol{u}\cdot\boldsymbol{e}_a)^2}{2c_s^4}-\frac{\boldsymbol{u}\cdot\boldsymbol{u}}{2c_s^2}\right) \tag{20}$$

The detailed temperature information obtained from Lattice Boltzmann direct numerical simulation method is utilized as the observed value to train the model, and the corresponding velocity and pressure are regarded as referenced values for model output.

3.2. Model architecture

To obtain the results reported in this work, each of the output variables (i.e., temperature φ, pressure p, x-direction component of velocity u, y-direction component of velocity v) was represented by an 8-layer deep neural network with 160 neurons per hidden layer. We considered three different network depths with 7, 8 and 9 layers, and three different neurons per hidden layer: 150, 160 and 170, respectively, resulting in a total of 9 combinations. We found that the chosen network structure (an 8-layer deep neural network with 160 neurons per hidden

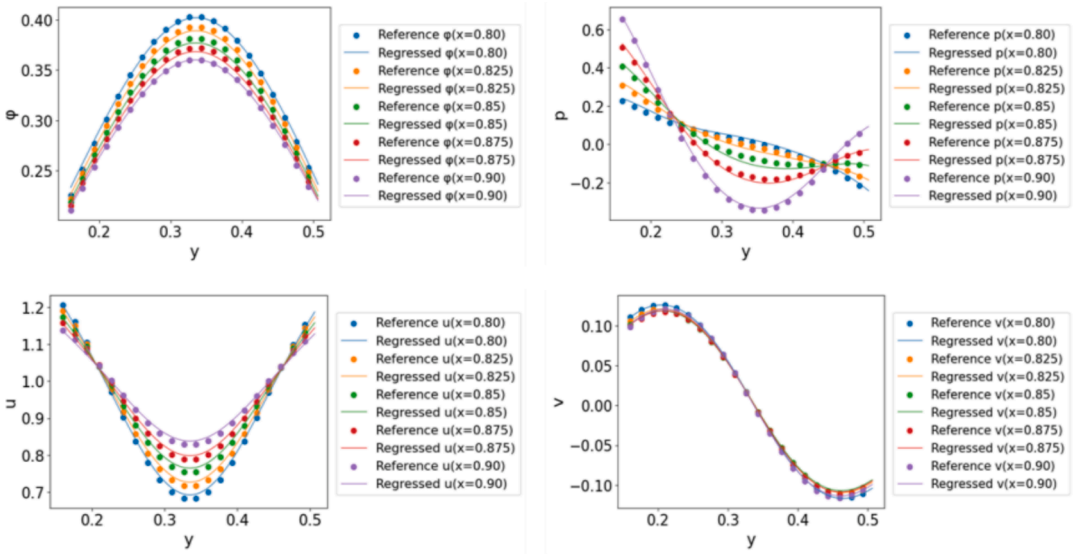

Fig. 22. Referenced and regressed φ, p, u, v for x = 0.80, 0.825, 0.85, 0.875 and 0.90 of forced convection heat transfer around a cylinder (Re = 10).

layer) had the smallest relative error of temperature, along with relatively small relative errors of velocity and pressure among the nine configurations. Therefore, we ultimately selected this structure. As for the activation functions, we chose the swish activation function explicitly defined by Eq. (21) (Raissi et al. 2020).

$$\text{swish}(x) := x\,\text{sigmoid}(x) = x/(1-\exp(-x)) \tag{21}$$

In general, the choice of a neural network's architecture, including layers/neurons and form of activation functions, is crucial and often remains an art that relies on one's ability to balance the trade-off between the expressivity and trainability of the neural network. The deeper and wider the network is, the more expressive it becomes and the more resource it requires for training. In this work, the architectures of the neural networks for all examples are consistent.

3.3. Performance qualification

To qualify the performance, we calculated the loss function Loss based on the mean squared error (MSE) described by Eq. (22), which accounts for deviation from physical laws and discrepancies in temperature observed value of the entire flow field, velocity and pressure observed values at boundary conditions. Additionally, we compute the relative error presented in Eq. (23) and absolute error described by Eq. (24), between network output values (i.e., temperature φ, pressure p, x-direction component of velocity u, y-direction component of velocity v) and the corresponding referenced values from Lattice Boltzmann direct numerical simulation. Here, y_i and $\widehat{y}_i$ represent referenced value and network regressed value respectively.

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \widehat{y}_i)^2 \tag{22}$$

$$\text{Relative error} = \sqrt{\frac{\frac{\sum_{i=1}^{N}(\widehat{Y}_i - Y_i)^2}{N}}{\frac{\sum_{i=1}^{N}(Y_i - \overline{Y}_i)^2}{N}}} \tag{23}$$

$$\text{Absolute error} = \widehat{Y}_i - Y_i \tag{24}$$

4. Validation examples

4.1. Natural convection in a square cavity

Natural convection in a square cavity has been extensively studied. The basic pattern is characterized by higher temperature on the left side, lower temperature on the right side, and the adiabatic conditions on the upper and lower sides (Bergman and Ramadhyani 1986; Perumal and Dass 2014). Some researchers have expanded their investigations to include situations with are baffles (Alouah et al. 2015; Frederick and Valencia 1989) or square block (Rahman et al. 2009; Shuja et al. 2009) in the cavity, or where the outer wall of the cavity is rectangular (Humphrey and Jacobs 1981), zigzag (Miyamoto et al. 1989), fin-shaped (Pandey et al. 2018), or crescent-shaped (Kousar et al. 2020).

At first we consider the basic problem and the describe the problem of natural convection in a square cavity as shown in Fig. 2 (Kang and Hassan 2011): the temperature on the left side is high, the temperature on the right side is low, the upper and lower sides are adiabatic. The boundary velocity is 0 and there is gravity in the cavity. The Rayleigh number (Ra) and Prandtl number (Pr) are 1000 and 0.71, respectively.

For natural convection in a square cavity, there are $Fx = 0$ in Eq. (14) and $\text{Fy} = g\beta(\varphi - \varphi_0)$ in Eq. (14), where $\varphi_0 = \frac{1}{2}(\varphi_{hot} + \varphi_{cold})$. Thus, it is known that temperature and velocity affect each other as indicted by the governing equations. Dimensionless the above Eqs. (13)–(16) as follows: $\text{X} = \frac{x}{L}$, $\text{Y} = \frac{y}{L}$, $\text{T} = \frac{t \cdot \nu_t}{L^2}$, $\text{U} = \frac{uL}{\nu_t}$, $\text{V} = \frac{\nu L}{\nu_t}$, $\text{P} = \frac{pL^2}{\rho \nu_t^2}$, and $\varphi = \frac{\varphi - \varphi_0}{\varphi_{hot} - \varphi_{cold}}$, where $\text{Pr} = \frac{\nu_f}{\nu_t}$, $\nu_f = u_{ref} L \sqrt{\frac{\text{Pr}}{Ra}}$, $Ra = \frac{g\beta(\varphi_{hot} - \varphi_{cold})L^3}{\nu_f \nu_t}$, $\Delta x = \frac{1}{L}$, and $\Delta t = \frac{u_{ref}}{L}$ in the lattice system to obtain new Eqs. (25)–(28) (Hussain and Hussein 2010; Pandey et al. 2018).

$$\frac{\delta U}{\delta X} + \frac{\delta V}{\delta Y} = 0 \tag{25}$$

$$\frac{\delta U}{\delta T} + \text{U}\frac{\delta U}{\delta X} + V\frac{\delta U}{\delta Y} = -\frac{\delta P}{\delta X} + Pr\left(\frac{\delta^2 U}{\delta X^2} + \frac{\delta^2 U}{\delta Y^2}\right) \tag{26}$$

$$\frac{\delta V}{\delta T} + \text{U}\frac{\delta V}{\delta X} + V\frac{\delta V}{\delta Y} = -\frac{\delta P}{\delta Y} + Pr\left(\frac{\delta^2 V}{\delta X^2} + \frac{\delta^2 V}{\delta Y^2}\right) + RaPr\varphi \tag{27}$$

$$\frac{\delta \varphi}{\delta T} + \text{U}\frac{\delta \varphi}{\delta X} + V\frac{\delta \varphi}{\delta Y} = \left(\frac{\delta^2 \varphi}{\delta X^2} + \frac{\delta^2 \varphi}{\delta Y^2}\right) \tag{28}$$

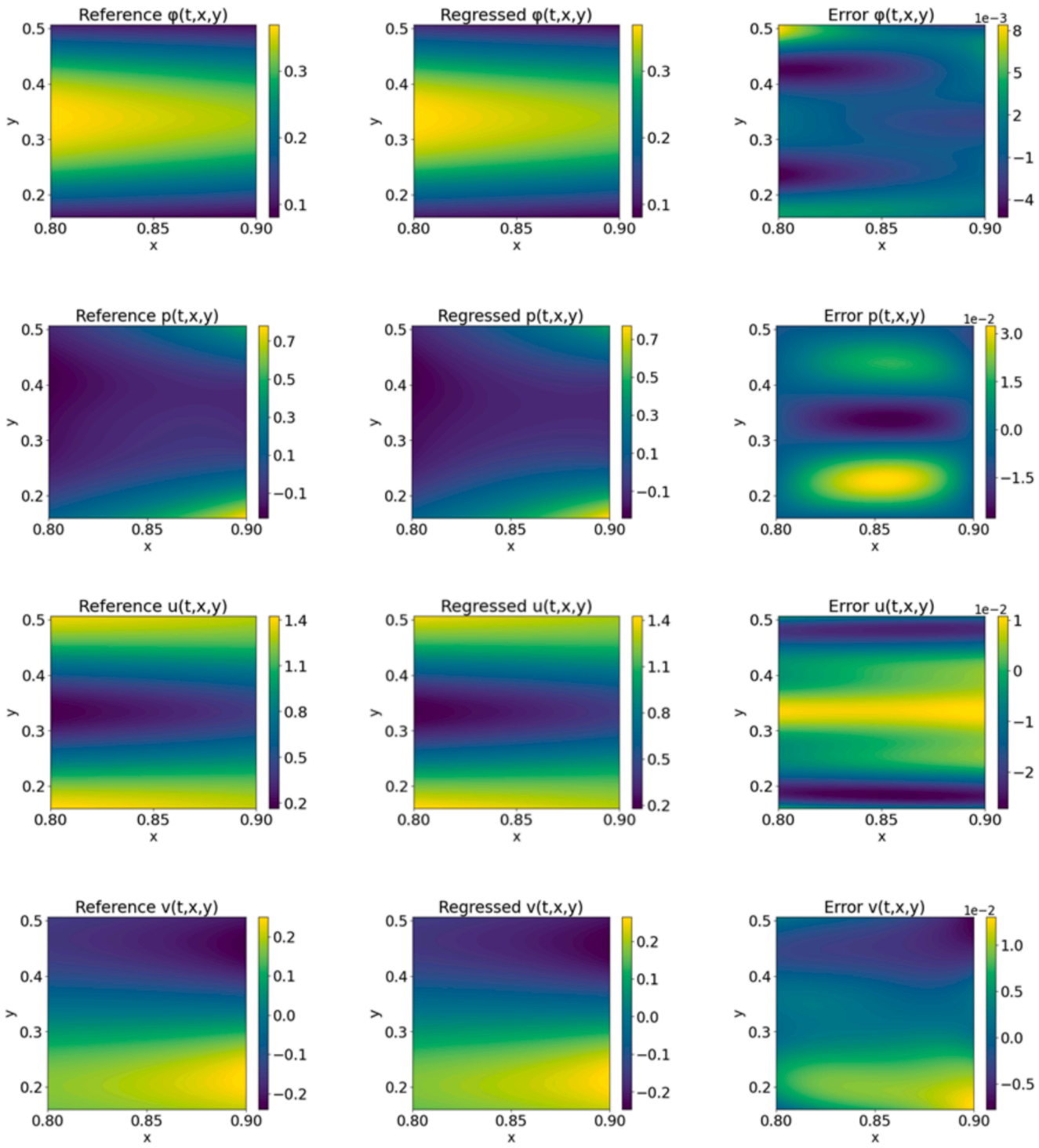

Fig. 23. Referenced and regressed φ, p, u, v, along with the corresponding absolute error for a specific selected region of forced convection heat transfer around a cylinder (Re = 40).

4.2. Flow around a cylinder in a square cavity

Based on the first example, we consider the flow around a cylinder in a square cavity. In the square cavity with a width L, a cylinder with a diameter D=0.4L is located at the center of the cavity, as shown in the Fig. 3. Cold and hot temperatures are applied to the vertical side walls and cylinder boundary, respectively. And adiabatic conditions are applied to the horizontal top and bottom walls. The experimental conditions of numerical simulation are Ra=1000 and Pr=0.71(Kang and Hassan 2011).

For this problem, the dimensionless equation form is the same as that for natural convection in a square cavity, as presented in Eqs. (25)–(28).

4.3. Two dimensional channel flow

In the channel flow case, the problem is described as shown in Fig. 4. Fluid enters the channel with a certain velocity and temperature on the left side, while the upper and lower walls are at low temperature with zero velocity. The far side of the channel is also at low temperature (Tang et al. 2003; Wang et al. 2006a, b). The Reynolds number (Re) and

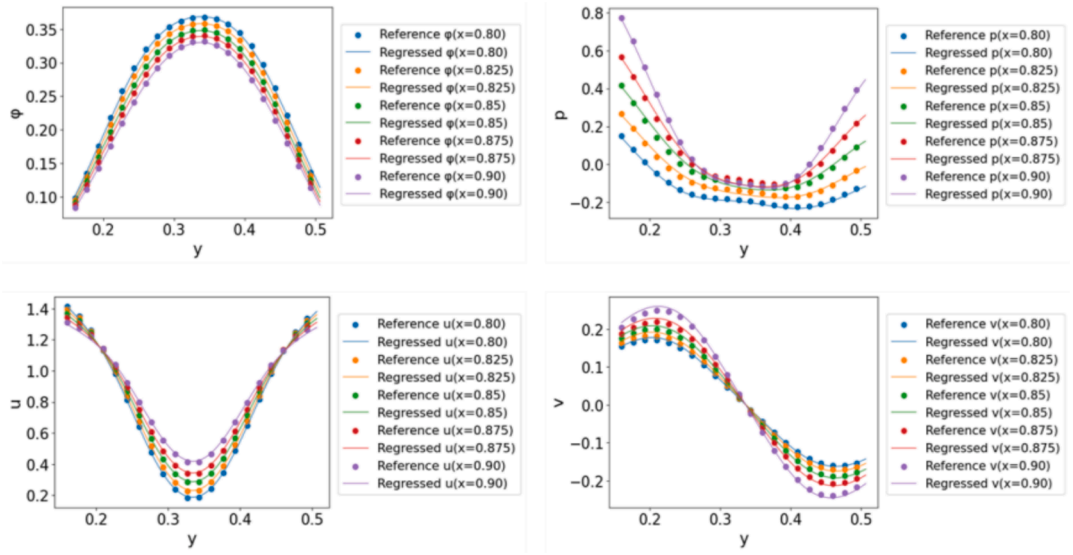

Fig. 24. Referenced and regressed φ, p, u, v for x = 0.80, 0.825, 0.85, 0.875 and 0.90 of forced convection heat transfer around a cylinder (Re = 40).

Pr are 10 and 0.7 respectively.

For two dimensional channel flow, there are $Fx = 0$ in Eq. (14) and Fy = 0 in Eq. (15), indicating that velocity affects temperature but temperature does not affect velocity. Dimensionless the above Equation (13)-(16) as follows: $X = \frac{x}{L}$, $Y = \frac{y}{L}$, $T = \frac{t \cdot u_{ref}}{L}$, $U = \frac{u}{u_{ref}}$, $V = \frac{v}{u_{ref}}$, $P = \frac{p}{\rho u_{ref}^2}$, and $\varphi = \frac{\varphi}{\varphi_{hot} - \varphi_{cold}}$, where $Pr = \frac{\nu_f}{\nu_t}$, $\nu_f = \frac{u_{ref} \cdot L}{Re}$, $\Delta x = \frac{1}{L}$, and $\Delta t = \frac{u_{ref}}{L}$ in the lattice system to get new momentum and energy Eqs. (29)–(31), while the continuum equation remains the same as Eq. (25) (Kousar et al. 2020):

$$\frac{\delta U}{\delta T} + U\frac{\delta U}{\delta X} + V\frac{\delta U}{\delta Y} = -\frac{\delta P}{\delta X} + \frac{1}{Re}\left(\frac{\delta^2 U}{\delta X^2} + \frac{\delta^2 U}{\delta Y^2}\right) \tag{29}$$

$$\frac{\delta V}{\delta T} + U\frac{\delta V}{\delta X} + V\frac{\delta V}{\delta Y} = -\frac{\delta P}{\delta Y} + \frac{1}{Re}\left(\frac{\delta^2 V}{\delta X^2} + \frac{\delta^2 V}{\delta Y^2}\right) \tag{30}$$

$$\frac{\delta \varphi}{\delta T} + U\frac{\delta \varphi}{\delta X} + V\frac{\delta \varphi}{\delta Y} = \frac{1}{RePr}\left(\frac{\delta^2 \varphi}{\delta X^2} + \frac{\delta^2 \varphi}{\delta Y^2}\right) \tag{31}$$

4.4. Forced convection heat transfer around a cylinder

Based on the third case, we consider the fourth case, forced convection heat transfer around a circular cylinder. Many researchers have studied this problem extensively (Cao et al. 2021; Haeri and Shrimpton 2013; Zhou et al. 2019). Fig. 5 illustrates the flow domain and boundary conditions for this case. The inlet, upper, and lower walls are located at a distance of L from the center of the cylinder, which has a diameter of 0.5 L. The outlet is positioned 2L downstream. Apart from the high temperature of the cylinder, the other four walls are maintained at a low temperature (Cao et al. 2021). The simulation condition for Re and Pr are 2(10, 40 and 140) and 0.71, respectively.

The dimensionless equation form for forced convection heat transfer around a cylinder is consistent with that of two dimensional channel flow, as described by Eqs. (25) and (29)–(31).

5. Results

We validate the idea that velocity and pressure of the flow field can be obtained from temperature distribution through four examples: natural convection in a square cavity, flow around a cylinder in a square cavity, two dimensional channel flow and forced convection heat transfer around a cylinder. We use three types of quantitative metrics to estimate the performance of PINN, that are Loss (summarized in Table 1), relative error (shown in the training process) and absolute error (shown in each figure providing snapshots of the performance of our PINNs at a given time). A single NVIDIA A 100 GPU is used for all our examples, and it takes approximately 0.15 s to complete a single step iteration. As shown in Table 1, the size of the sampling dataset for the spatial coordinate x and y is directly related to the grid resolution of LBM, referred to as the "spatial resolution". The size of the time coordinate t is randomly selected and is represented by "training snapshots", which indicate the number of snapshots used for training. Initially, we chose 201 training snapshots based on the study by Raissi et al. (Raissi et al. 2020). However, we later found that reducing t could help decrease the training time. Thus, we selected either 101 or 28 training snapshots. The "time step interval" refers to the sampling interval for time step in LBM, which is commonly set to 1000 for many cases. However, for complex flow problem, it is set to 1 to improve the prediction accuracy of PINN. The total number of sampling collection points is the product of x, y and t, which reflects the size of the temperature data. For example, for forced convection heat transfer around a cylinder (Re = 2), the size of the temperature data used for training is 1,680,000 (the product of x = 300, y = 200, and t = 28) with a time step interval of 1.

5.1. Natural convection in a square cavity

For natural convection in a square cavity, sampling is performed every 1000 steps of Lattice Boltzmann simulation. 201 time segments from the 198000th to 398000th interval points in 800 × 800 grid are selected to form a data set. As a result, the sampling dataset size for the spatial coordinate x, y and time coordinate t is 800, 800 and 201 respectively with a time step interval of 1000. And the size of the temperature data used for training is 128,640,000 (the product of x = 800, y = 800, and t = 201). The selected research area, $[0.25, 0.75] \times [0.25, 0.75]$, is arbitrarily chosen and located at the center of the cavity. The entire cavity area is represented as $[0, 1] \times [0, 1]$.

From Fig. 6 and Fig. 7, we can see that Loss and relative error keep decreasing as the number of iterations increases until it stabilizes after 10^5 iterations. These prove that the output of the network are close to the referenced values. For a specific time period, Fig. 8 shows referenced and regressed φ, p, u, v, along with the corresponding absolute error for selected area. The regressed velocity and pressure, derived from the network without additional initial information, are consistent with referenced flow feature. Across the entire selected research area (401×401 scatter point), maximum absolute error of u, v is on the order

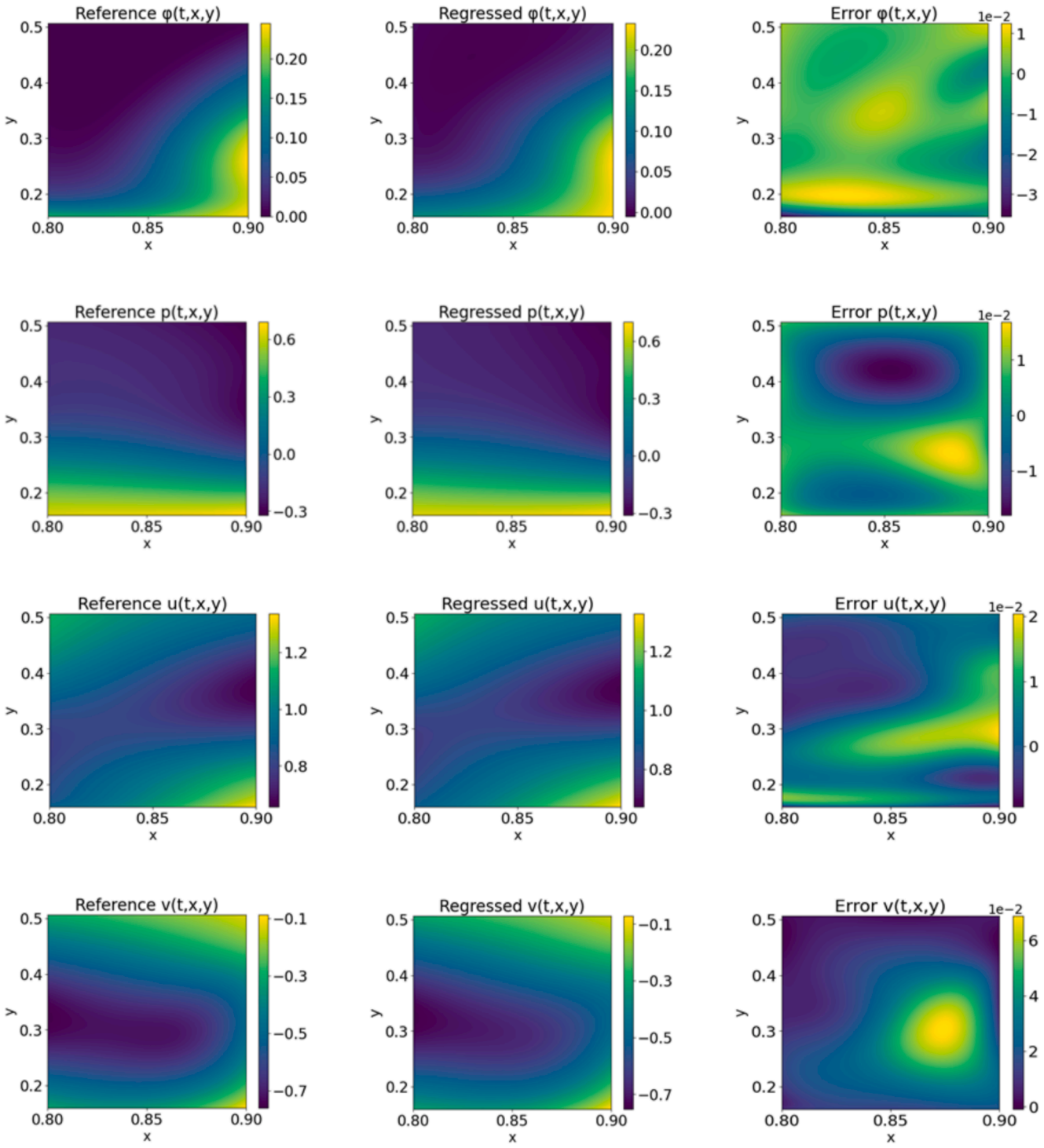

Fig. 25. Referenced and regressed φ, p, u, v, along with the corresponding absolute error for a specific selected region of forced convection heat transfer around a cylinder (Re = 140).

of 0.01 and for p it is on the order of 0.1. We can see a more straightforward comparison between the referenced and regressed φ, p, u, v from Fig. 9, where y ranges from 0.25 to 0.75 along the line while x are chosen as 0.3, 0.4, 0.5, 0.6 and 0.7 at different positions. It can be observed that the regressed φ, p, u, v are consistent with referenced values. All of the above prove the precise accuracy of the regressed φ, p, u, v.

5.2. Flow around a cylinder in a square cavity

Although the governing equations of flow around a cylinder in a square cavity remains the same as those for natural convection in a square cavity, the complex boundary created by the cylinder results in a more intricate flow field.

For flow around a cylinder in a square cavity, sampling is performed every 1000 steps of Lattice Boltzmann simulation. 101 time segments from the 243000th to 343000th interval points in 800×800 grid are selected to form a data set. As a result, the sampling dataset size for the spatial coordinate x, y and time coordinate t is 800, 800 and 101 respectively with a time step interval of 1000. And the size of the temperature data used for training is 64,640,000 (the product of x = 800, y

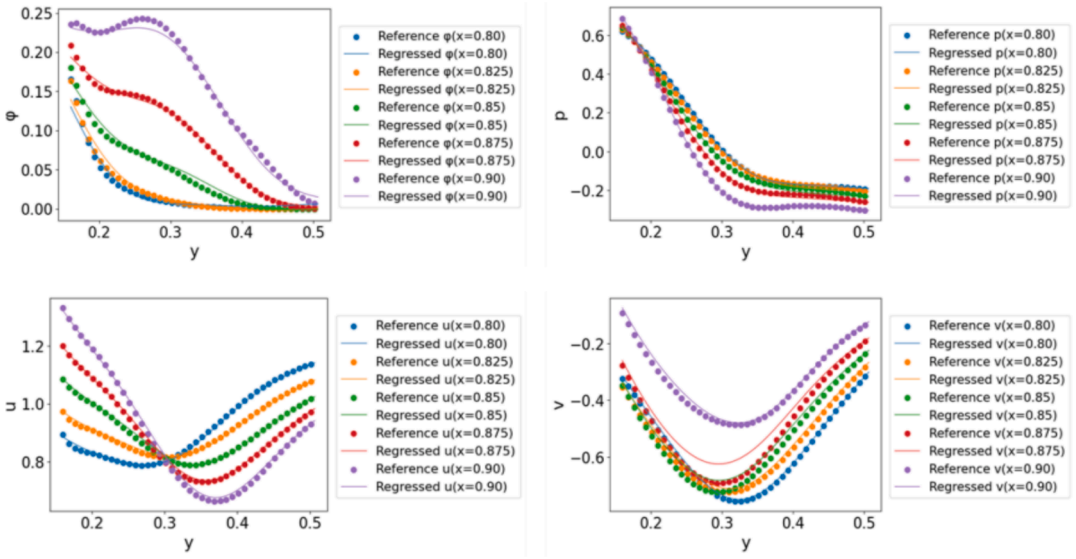

Fig. 26. Referenced and regressed φ, p, u, v for x = 0.80, 0.825, 0.85, 0.875 and 0.90 of forced convection heat transfer around a cylinder (Re = 140).

= 800, and t = 101). The arbitrarily selected research area is [0.75, 0.95] × [0.75, 0.95], at the upper right corner of the cavity. Similarly, the entire cavity area is depicted as $[0,1] \times [0,1]$.

From Fig. 10 and Fig. 11, it can be observed that the number of iterations for the flow around a cylinder in a square cavity is on the order of 10^6, which is larger than that for natural convection in a square cavity, which requires around 10^5 iterations. This difference can be attribute to the fact that the more complex the problem, the more iterations the algorithm needs to convergence. Fig. 11 shows that after 10^6 iterations, the relative error of φ tends to stabilize, which is an important indicator of convergence. More detailed results can be found in Fig. 12 and Fig. 13.

From Fig. 12, the maximum absolute error for p, u, v in the entire selected research area (161×161 scatter point) are all on the order of 0.01, while the maximum absolute error for φ only reaches 0.1. To compute the difference between the predicted and observed pressure fields, we subtract the spatial average pressure from both predicted and exact fields. This step was necessary, because it is the differential pressure not the absolute pressure that appears in the governing equations (Raissi et al., 2019b, 2020). We found that there is no need for pressure post-processing in the case of natural convection in a square cavity. From Fig. 13, an interesting result can be observed: the accuracy of temperature prediction is less than of velocity and pressure, even though the temperature field is observed as input to the network. The reason may be that, unlike traditional pure data-driven machine learning algorithm where the loss function only accounts for the data error between observed values and predictions, the loss function for PINN includes errors from partial differential equation and boundary condition to enforce physical laws, in addition to data errors. To some extent, minimizing the loss function during the iteration process may limit the network's ability to perfectly approximate the observed values, but it improves the prediction accuracy for physical quantities other than the observed values. In general, velocity and pressure can be derived from temperature distribution.

5.3. Two dimensional channel flow

For two dimensional channel flow, where velocity affects temperature but temperature does not affect velocity, sampling is conducted every 1000 steps of Lattice Boltzmann simulation. 201 time segments from the 280000th to 480000th interval points in 1600 × 1600 grid are collected to form a data set. As a result, the sampling dataset size for the spatial coordinate x, y and time coordinate t is1600, 1600 and 201 respectively with a time step interval of 1000. And the size of the temperature data used for training is 514,560,000 (the product of x = 1600, y = 1600, and t = 201). The selected research area, [0.25, 0.75] × [0.25, 0.75], is arbitrarily chosen and located at the center of the channel. The entire channel area is represented as $[0,1] \times [0,1]$.

From Fig. 14 and Fig. 15, we can observe that two-dimensional channel flow converges after approximately 10^4 iterations, which is fewer than 10^5 iterations required for natural convection in a square cavity and 10^6 iterations needed for flow around a cylinder in a square cavity. In Fig. 16, for a given time, the maximum absolute error of φ, p, u over the entire selected research area (801×801 scatter point) is on the order of 0.001, while the maximum absolute error of v even reaches 0.0001. Fig. 17 provides a more direct comparison between the referenced and regressed φ, p, u, v. Here, y ranges from 0.25 to 0.75 along axis, while x are selected as 0.3, 0.4, 0.5, 0.6 and 0.7 at different positions. It can be observed that all regressed φ, p, u, v match the corresponding referenced values well.

5.4. Forced convection heat transfer around a cylinder

Based on the third example of two-dimensional channel flow, we delved into a detailed discussion about forced convection heat transfer around a cylinder. We explored different Reynolds numbers, including 2, 10, 40 and 140 for various flow conditions. As is well known, when the Reynolds number is below $Re \approx 45$, the flow around a circular cylinder remains steady over time. However, at Reynolds number above $Re \approx 45$, the flow becomes unsteady (Cao et al. 2021). Fig. 18 illustrates the flow and temperature distribution at different Reynolds numbers for forced convection heat transfer around a cylinder (Re= 2, 10, 40 and 140). It is evident that when the Reynolds number is 140, the entire flow field changes drastically. As show in Table 1, for Reynolds number of 2, 10 and 40, the sampling dataset size for the spatial coordinate x, y and time coordinate t is 300, 200 and 28 respectively with a time step interval of 1. And the size of the temperature data used for training is 1,680,000 (the product of x = 300, y = 200, and t = 28). As for Reynolds number of 140, the sampling dataset size for the spatial coordinate x and y is 600 and 400 respectively and the corresponding size of the temperature data is 6,720,000.

The governing equations for forced convection heat transfer around a cylinder are in the same form as those for two dimensional channel flow, as described in Equation (25)(29)(30)(31). The arbitrarily selected training research area is [0.80, 0.90] × [0.24, 0.76] within the rectangular region represented as $[0,1] \times [0,1]$. Fig. 19, Fig. 21, Fig. 23 and

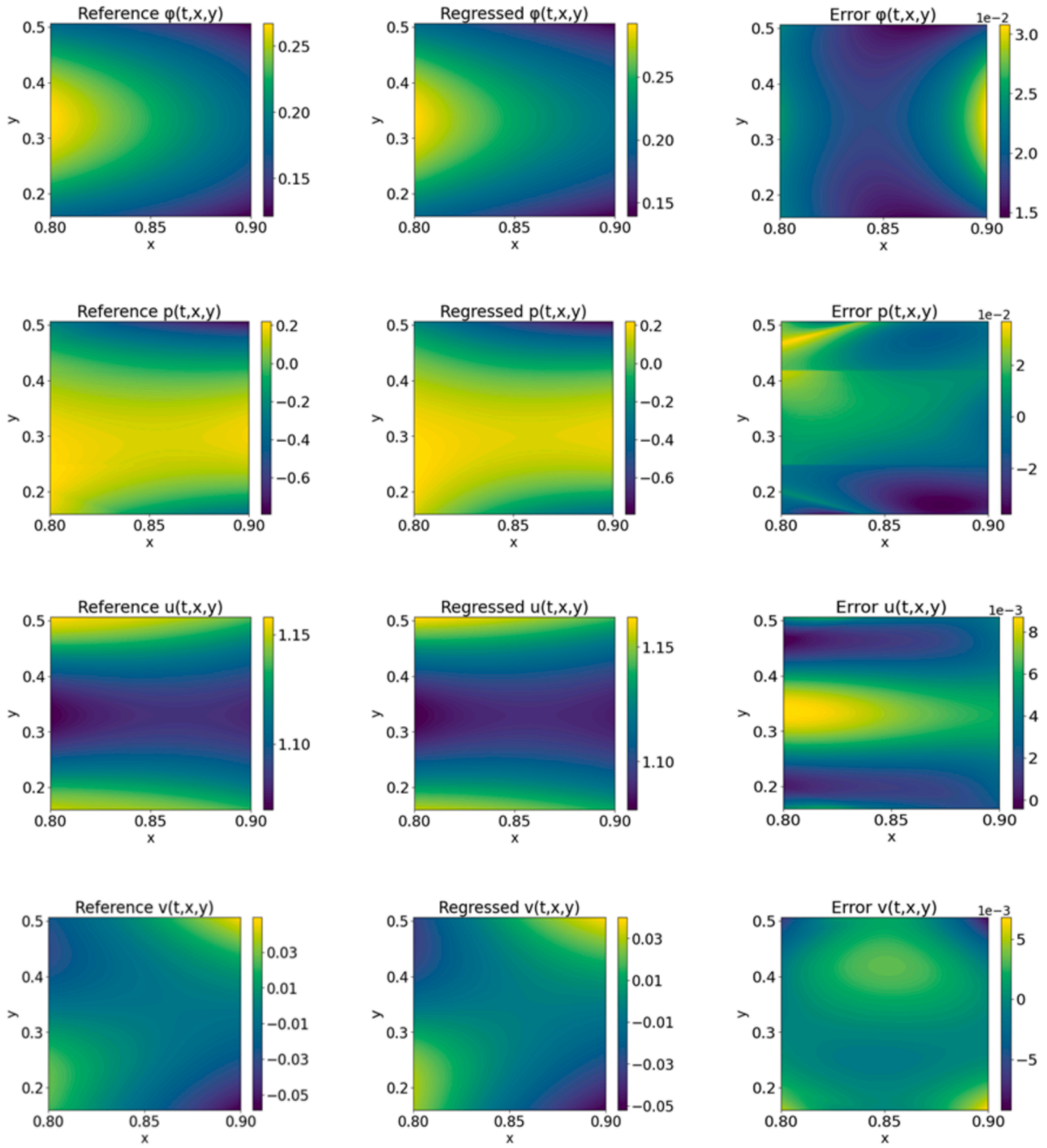

Fig. 27. Referenced and regressed φ, p, u, v, along with absolute error, for $[0.80, 0.90] \times [0.24, 0.76]$ region of forced convection heat transfer around a cylinder with 600×400 grid (Re $= 2$, model trained at area of $[0.80, 0.90] \times [0.24, 0.76]$ through 300×200 grid).

Fig. 25 depict the referenced and regressed p, u, v, along with the corresponding absolute error, for the selected area of forced convection heat transfer around a cylinder at different Re of 2, 10, 40 and 140, respectively. The maximum absolute error for all cases is on the order of 10^{-2}, with many even reaching the order of 10^{-3}. Fig. 20, Fig. 22, Fig. 24 and Fig. 26 shows the corresponding referenced and regressed φ, p, u, v for x = 0.80, 0.825, 0.85, 0.875 and 0.90 of forced convection heat transfer around a cylinder. It can be observed that all regressed φ, p, u, v are consistent with the corresponding referenced values.

Overall, velocity and pressure can be derived from temperature distribution for different Reynolds numbers in the forced convection heat transfer around a cylinder.

For Re=2, the model is trained at $[0.80, 0.90] \times [0.24, 0.76]$ of the rectangular region through 300×200 grid. For in depth research, we directly use the well trained model to predict the fluid flow temperature, velocity and pressure for $[0.80, 0.90] \times [0.24, 0.76]$ and $[0.70, 0.80] \times [0.24, 0.76]$ area at 600×400 grid. The former means small sample temperature data can get more precise velocity and pressure information. The latter has a great influence for velocity measurement in a region that lacks prior information is and is far from the trained region.

Regarding super-resolution of the same region, Fig. 27 and Fig. 28 demonstrate that temperature, velocity and pressure can be directly

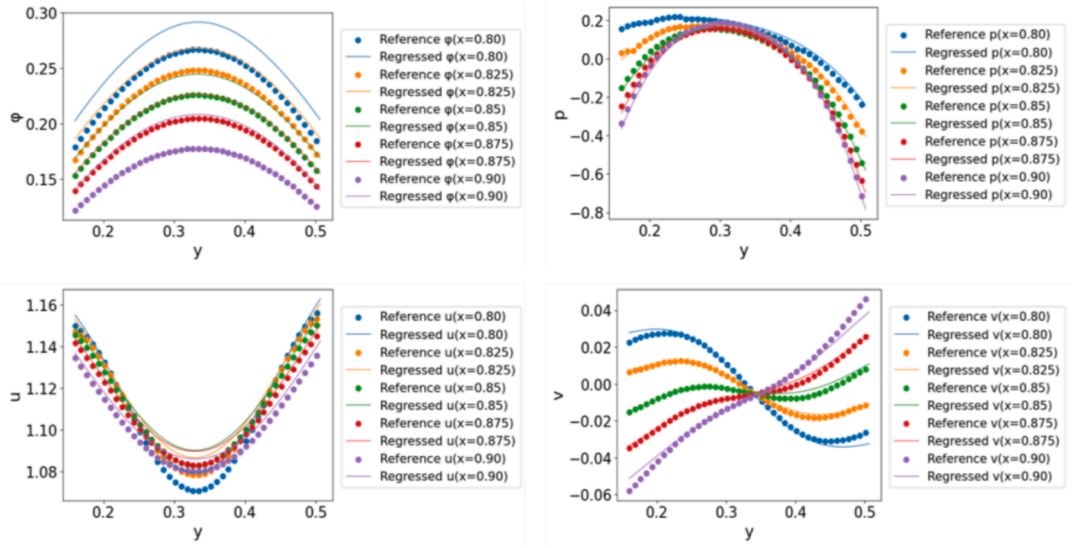

Fig. 28. Referenced and regressed φ, p, u, v for x $= 0.80, 0.825, 0.85, 0.875$ and 0.90 of forced convection heat transfer around a cylinder with 600×400 grid (Re $=$ 2, model trained at $[0.80, 0.90] \times [0.24, 0.76]$ through 300×200 grid).

obtained through the spatiotemporal position coordinate input of the network, with the regressed values closely matching the corresponding referenced values. However, for region outside the trained region, the predicted performances of velocity u and v are acceptable, while pressure and temperature results are not excellent, as shown in Fig. 29 and Fig. 30. This can be attributed to the fact that the well-trained model only contains temperature information with $[0.80, 0.90] \times [0.24, 0.76]$ region and governing equations, without including any information from the $[0.70, 0.80] \times [0.24, 0.76]$ region.

5.5. Accuracy and consumption time comparison between LBM and PINN

In this study, LBM solves the forward problem, while PINN solves the inverse problem. It is nearly impossible to obtain analytical solutions or simultaneous experimental values for temperature, velocity, and pressure across the entire field for all cases considered in this work. Thus, the LBM results are regarded as reference values for PINN. The temperature field data from LBM are regarded as observed values used for PINN training. The output variable of PINN including temperature, velocity, and pressure, are compared with the corresponding values from LBM to evaluate the accuracy of the PINN method.

For steady-state problems, the solution time of DNS is determined by the convergence condition where the error between previous and subsequent iterations is less than 10^{-9}, while the training time for PINN is custom-set by us (because the value of loss when it converges is uncertain shown in Table 1). Additionally, the LBM code and PINN code run on the CPU and GPU respectively, which are not comparable as they depend on different hardware. However, we can roughly compare the computational time between LBM and PINN. For example, in the forced convection heat transfer around a cylinder, as shown in Table 2, the PINN training time is slow (about 12 h), but the prediction time is fast (less than 1 min). The velocity, pressure, and temperature of the flow field can be quickly obtained without retraining, while LBM requires resolving (about 1.37 h). Therefore, when more than 9 tests are required, the efficiency of PINN surpasses that of LBM. We aim to explore the feasibility and generality of the PINN method for obtaining velocity and pressure based on the temperature field, so we focus primarily on the accuracy of PINN solution. As shown in Table 3, the accuracy of the temperature, velocity and pressure predicted by the network is very high (the relative error of velocity and pressure is about 10^{-2}) under both training and testing conditions.

In terms of comparison between different cases, as seen in Table 3, the prediction errors for the cases in presence of a cylinder are generally larger than those for cases in the absence of a cylinder (e.g., comparing flow around a cylinder in a square cavity with natural convection in a square cavity, as well as forced convection heat transfer around a cylinder with two-dimensional channel flow). Meanwhile, the prediction errors for the high Reynolds number case are larger than those for the intermediate Reynolds number cases (e.g., comparing Reynolds number of 140 in forced convection heat transfer around a cylinder with Reynolds numbers of 10 and 40 in similar conditions). There is indeed an increase in the numerical error at the walls and under high Reynolds number flow condition in the LBM simulation and the above results indicate that the prediction accuracy of PINN is closely related to the precision of observed values. Additionally, the prediction errors for the test set are larger than those for the training set (e.g., comparing Reynolds number of 2 in forced convection heat transfer around a cylinder outside the trained region with that in the trained region). This can be attributed to the difference in data distribution between the training set and the test set.

We had trained on the entire region for all cases in our preliminary investigation. We found that the network output accuracy for temperature, velocity, and pressure in the entire region for natural convection in a square cavity and two-dimensional channel flow were quite satisfactory. Fig. 31 shows the results for two-dimensional channel flow. However, the accuracy for other cases, particularly in the presence of a cylinder, was not as good as expected. Fig. 32 shows the results for Reynolds numbers of 40 in forced convection heat transfer around a cylinder. Therefore, for all cases studied in this work, including natural convection in a square cavity, flow around a cylinder in a square cavity, two-dimensional channel flow, as well as different Reynolds numbers of 2, 10, 40 and 140 in forced convection heat transfer around a cylinder, we have chosen a local arbitrary area far from the walls as the study region to ensure the generalizability of our proposed method and the error distribution in the local trained region is random.

The underlying reason for the different accuracy at different tested cases might be that, our original data for all these studies were obtained by direct numerical simulations with LBM. In this work, we use the results of the LBM as observations. The advantage of this method is that it does not require grid encryption at the wall, thereby reducing a lot of computing time and space, but the number of sampling data near the wall is relatively small, which leads to the relatively larger numerical errors at the wall and makes it hard to accurately learn enough physical information near the boundary for PINN prediction. Therefore, the

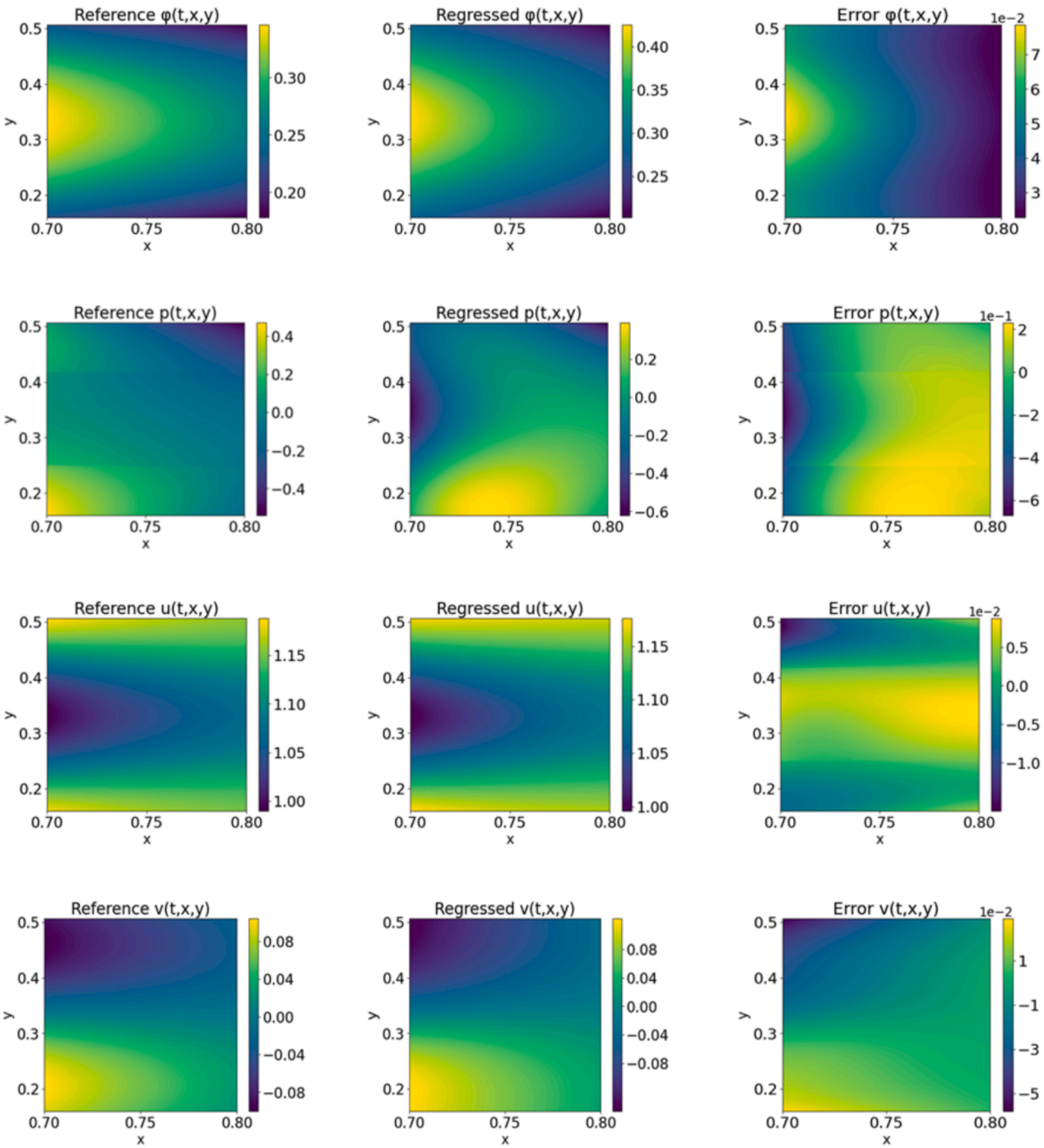

Fig. 29. Referenced and regressed φ, p, u, v, along with absolute error for $[0.70, 0.80] \times [0.24, 0.76]$ region of forced convection heat transfer around a cylinder with 600×400 grid ($\mathrm{Re} = 2$, model trained at $[0.80, 0.90] \times [0.24, 0.76]$ through 300×200 grid).

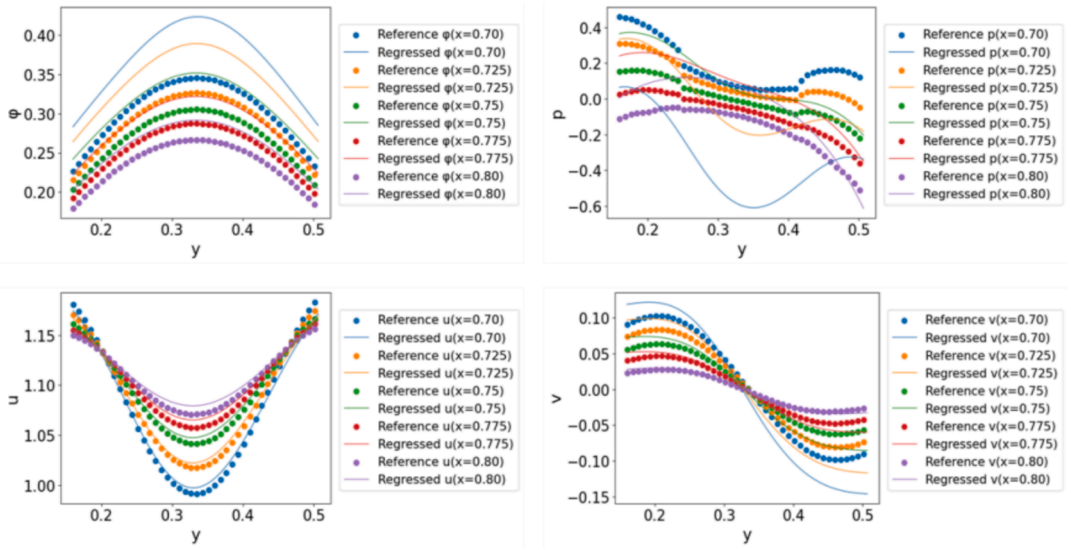

Fig. 30. Reference and regressed φ, p, u, v for x $= 0.70, 0.725, 0.75, 0.775$ and 0.80 of forced convection heat transfer around a cylinder with 600×400 grid (Re $= 2$, model trained at $[0.80, 0.90] \times [0.24, 0.76]$ through 300×200 grid).

Table 2
Comparison of LBM and PINN consumption time in forced convection heat transfer around a cylinder.

	Case	Time	
		DNS	PINN
Train	Forced convection heat transfer around a cylinder (Re = 2)	0.21 h	12 h
Train	Forced convection heat transfer around a cylinder (Re = 10)	0.17 h	12 h
Train	Forced convection heat transfer around a cylinder (Re = 40)	1.17 h	12 h
Train	Forced convection heat transfer around a cylinder (Re = 140)	~0.71	12 h
Test	Re = 2 (super resolution in the trained area) in forced convection heat transfer around a cylinder	1.37 h	<1 min
Test	Re = 2 (super resolution out of the trained area) in forced convection heat transfer around a cylinder	1.37 h	<1 min

Table 3
Temperature, velocity and pressure relative error obtained in different flow problems and conditions.

	Case	φ	u	v	p
Train	Natural convecition in a square cavity	7.25×10^{-2}	6.76×10^{-3}	2.21×10^{-2}	2.79×10^{-2}
Train	Flow around a cylinder in a square	9.45×10^{-1}	7.99×10^{-2}	4.01×10^{-2}	7.48×10^{-1}
Train	Two-dimensional channel flow	5.49×10^{-3}	3.29×10^{-3}	2.53×10^{-3}	1.32×10^{-2}
Train	Forced convection heat transfer around a cylinder (Re = 2)	9.52×10^{-2}	1.05×10^{-1}	1.66×10^{-1}	4.35×10^{-2}
Train	Forced convection heat transfer around a cylinder (Re = 10)	5.24×10^{-2}	5.99×10^{-2}	1.82×10^{-2}	1.11×10^{-1}
Train	Forced convection heat transfer around a cylinder (Re = 40)	2.50×10^{-2}	3.06×10^{-2}	3.53×10^{-2}	7.81×10^{-2}
Train	Forced convection heat transfer around a cylinder (Re = 140)	7.55×10^{-2}	6.68×10^{-2}	1.59×10^{-1}	3.43×10^{-2}
Test	Re = 2 (super resolution in the trained area)	6.41×10^{-1}	2.08×10^{-1}	1.11×10^{-1}	8.60×10^{-2}
Test	Re = 2 (super resolution out of the trained area)	1.20×10^{0}	1.15×10^{-1}	2.95×10^{-1}	9.84×10^{-1}

results we give don't consider the vicinity of the boundary and the training research area for the PINN is [0.80, 0.90] × [0.24, 0.76] far from the cylinder within the rectangular region represented as [0, 1] × [0, 1] in Section 5.4. However, when there are enough observations near the wall, we found that encrypting more mesh near the cylinder can improve prediction accuracy, as shown in Fig. 33 for Reynolds numbers of 40 in forced convection heat transfer around a cylinder with 250,000 data points encrypted near the cylinder. Overall, the general boundary encryption physics-informed neural network method will be the focus of our next research work.

6. Conclusions

In this work, a general Physics-Informed Neural Network approach for deriving fluid flow fields from temperature distribution is proposed. The feasibility of using the temperature distribution to predict the velocity and pressure of the flow field is verified in three cases: natural convection in a square cavity, flow around a cylinder in a square cavity, and two-dimensional channel flow.

In natural convection of the square cavity, where velocity and temperature interact, 10^5 iterations are required. The maximum absolute error of x-direction component of velocity u and y-direction component of velocity v is on the order of 10^{-2}, and the maximum absolute error of pressure p is 10^{-1} in the selected research area (401×401 scatter point). For flow around a cylinder in a square cavity, because temperature and velocity are coupled and there is a cylinder, it takes 10^6 iterations for the loss to converge. The maximum absolute error of p, u and v all reach the order of 10^{-2} in the selected research area (161×161 scatter point). After 10^4 iterations of the two-dimensional channel flow, where only velocity affects temperature but temperature does not affect velocity, the loss no longer changes. The maximum absolute error of pressure p and x-direction component of velocity u in the selected research area (801×801 scatter point) reaches the order of 10^{-3}, and the maximum absolute error of y-direction component of velocity v even reaches 10^{-4}.

We then examined the flow with forced convection heat transfer around a cylinder. Different Reynolds number 2, 10, 40 and 140 flow conditions are considered to obtain velocity and pressure information based on fluid temperature distribution. The regressed values are almost consistent with the reference values, with maximum absolute error reaching the order of 10^{-2} in the selected area and even many reaching the order of 10^{-3}. For super-resolution prediction of the same region as the trained region, temperature, velocity and pressure can be directly

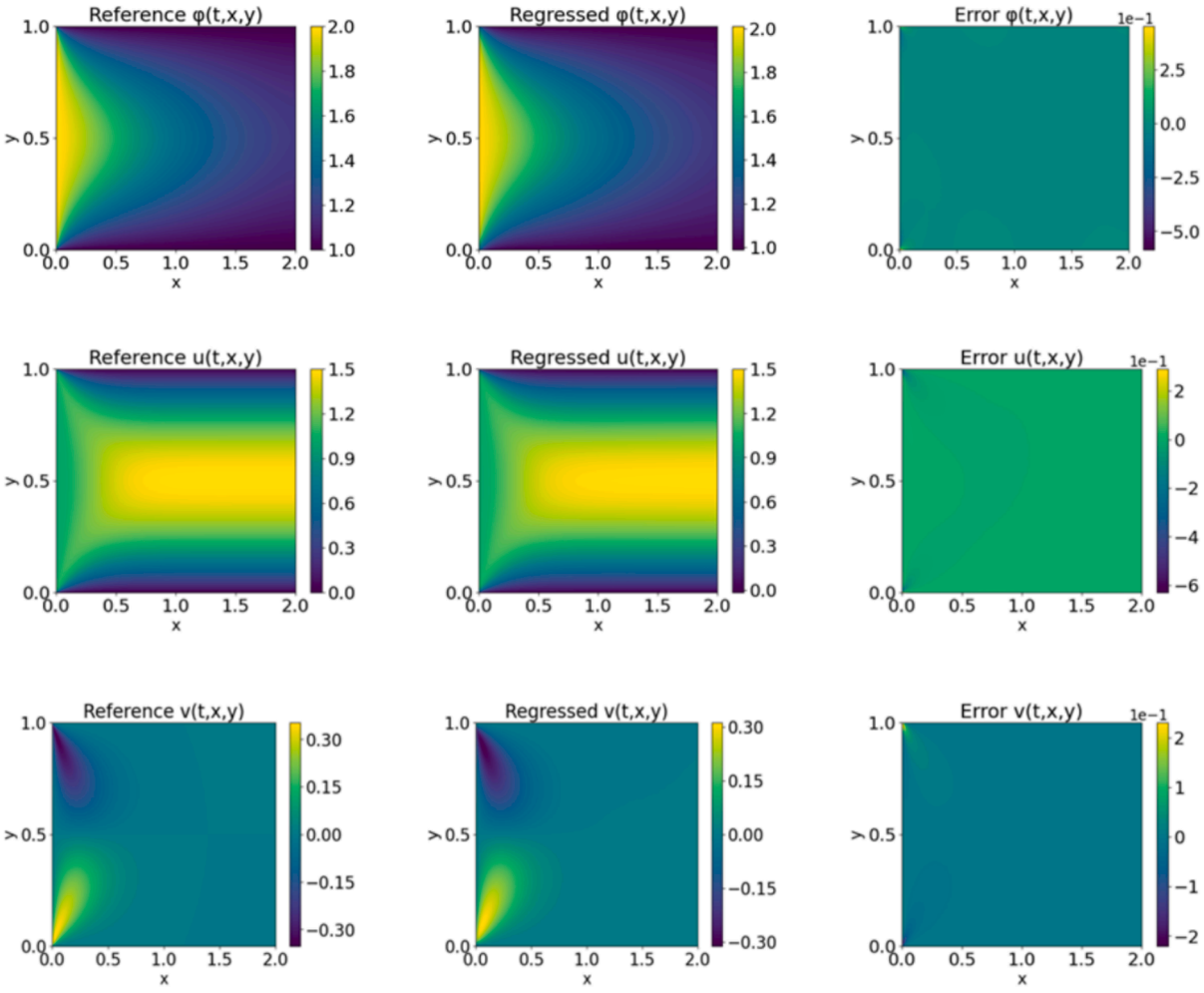

Fig. 31. Referenced and regressed φ, u, v along with the corresponding absolute error for the entire region of two-dimensional channel flow.

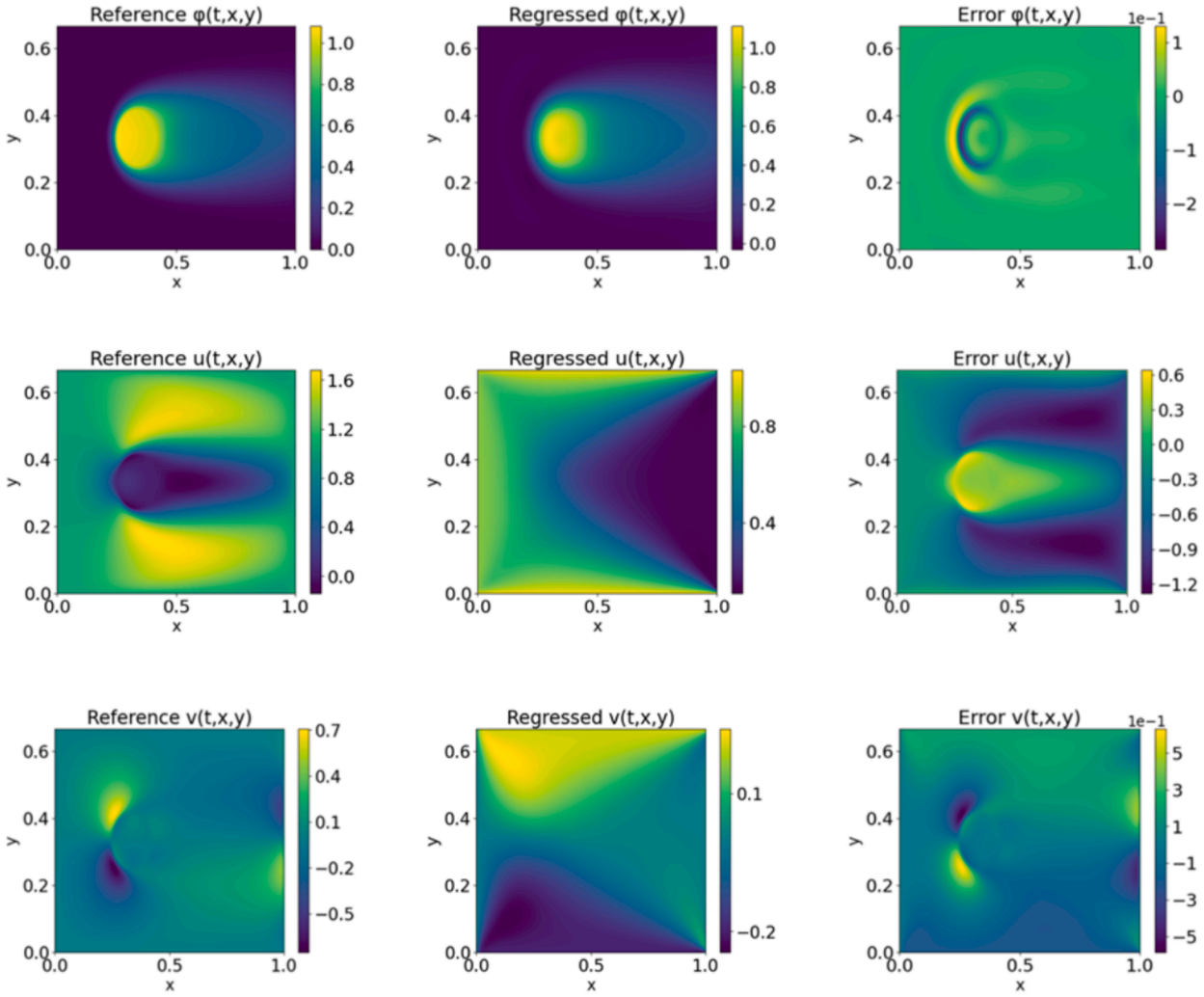

Fig. 32. Referenced and regressed φ, u, v along with the corresponding absolute error for the entire region of forced convection heat transfer around a cylinder (Re = 40).

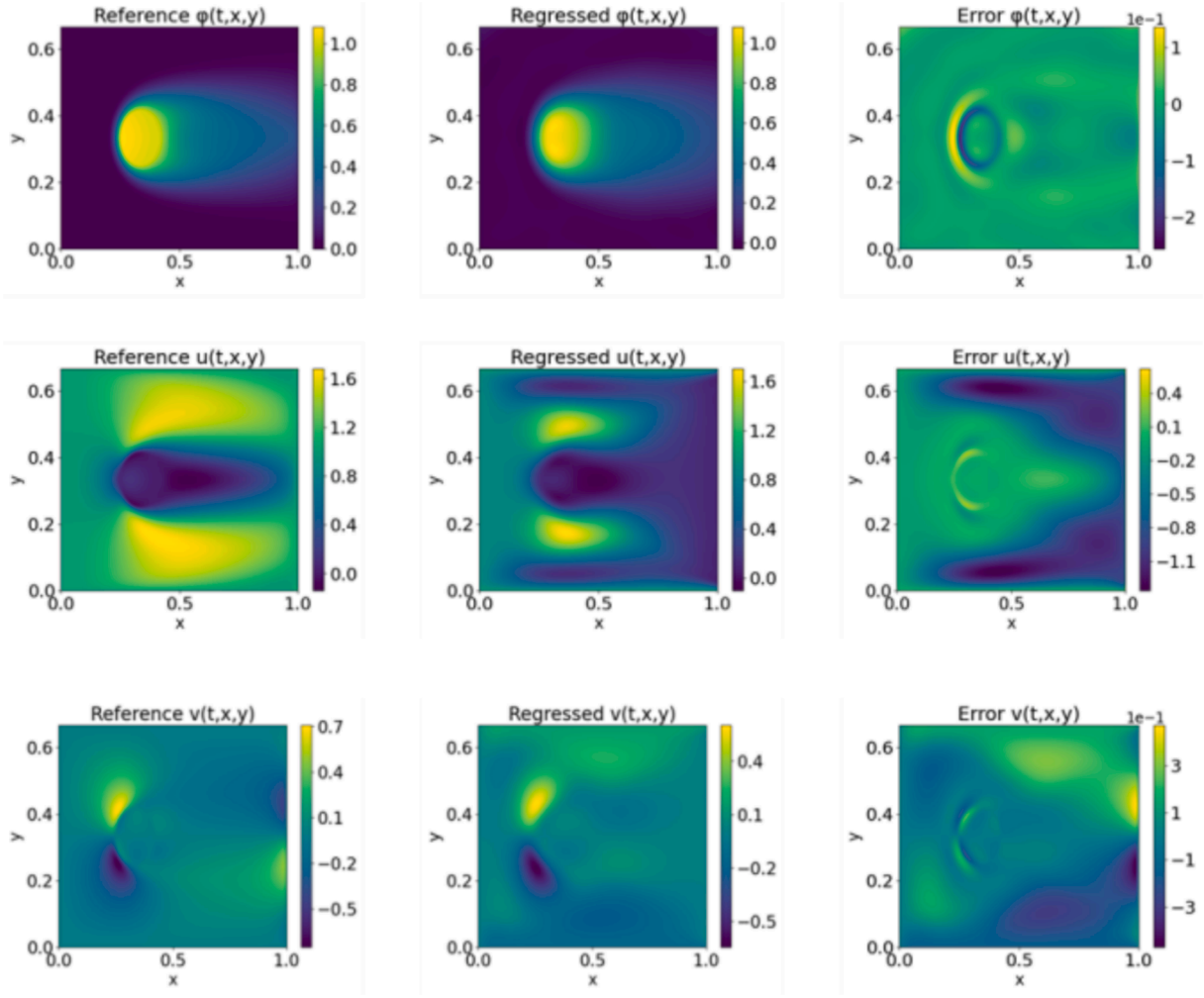

Fig. 33. Referenced and regressed $\varphi, \mathrm{u}, \mathrm{v}$ along with the corresponding absolute error for the entire region of forced convection heat transfer around a cylinder with 250,000 data points encrypted near the cylinder (Re = 40).

obtain through the spatiotemporal position coordinates of network input. However, for the regions outside the trained region, the predicted performance of velocity u and v is acceptable, while pressure and temperature results are not as accurate.

For the boundary conditions, we considered constant flow boundaries, rebound boundaries, and wall boundaries, which covered most common boundary conditions. Our method is also applicable to non-steady-state problems (e.g. Reynolds numbers of 140 in forced convection heat transfer around a cylinder) and can acquire the velocity and pressure fields based on the temperature field. Therefore, more complex boundaries or time-related boundaries are theoretically applicable.

The results provide a theoretical basis for future experiments and offer a new approach to acquiring velocity and pressure in heat transfer problem where temperature measurements are available but the initial conditions of velocity and pressure may be unknown. Although velocity and pressure boundary condition information are still required as temperature alone does not contain enough information about the flow under complex flow conditions to determine unique velocity and pressure, the measurement cost is greatly reduced.

CRediT authorship contribution statement

Cheng Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chenggong Li:** Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization. **Xue Li:** Writing – review & editing, Visualization, Validation, Resources, Funding acquisition, Formal analysis, Conceptualization. **Mao Ye:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition, Formal analysis, Conceptualization. **Zhongmin Liu:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

We have shared the link to our data in the manuscript. The numerical simulation data for natural convection in a square cavity and the corresponding PINN model predictions presented in section 5.1 are available in Data S1. The results presented in Figures 6-9 can be reproduced using this data with three scripts provided in Data S1. Furthermore, the numerical simulation data for flow around a cylinder in a square cavity

and the corresponding PINN model predictions presented in section 5.2 are available in Data S2. The results presented in Figures 10-13 can be reproduced using this data with the script provided in Data S2. The numerical simulation data and corresponding PINN model predictions for two-dimensional channel flow presented in section 5.3 are available in Data S3.This data supports the results presented in Figure 14-17. The numerical simulation data and corresponding PINN model predictions presented in section 5.4 are available in Data S4. The results presented in Figures 19-30 can be reproduced using this data with three scripts provided in Data S4. Data S1, Data S2, Data S3 and Data S4 can be downloaded from Google Drive at https://drive.google.com/drive/folders/1IJXeBkisqj_ONcaMXWV5zfSfik4mhZBw?usp=drive_link.

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